Closing the Three-Gap Framework: A Systematic Review of Evaluation, Governance, and Implementation Challenges in Clinical AI Deployment *

Master's Thesis

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Abstract

Despite rapid growth in clinical AI research, deployment outcomes remain poorly understood because evaluation, governance, and implementation are studied in isolation across disconnected literatures. This review proposes a three-gap framework as its central contribution, identifying three interdependent structural deficits: an **ecological validity gap** between controlled validation and clinical reality, a **governance capacity gap** between regulatory assumptions and health system readiness, and a **pilot trap** reflecting systematic pilot-to-routine transition failure. A systematic search of PubMed, Semantic Scholar, arXiv, and DBLP (2020–2026) yielded 311 papers, of which 60 were selected for structured content analysis based on methodological relevance and thematic centrality. Empirical analysis identifies a consistent pattern linking ecological validity failure to pilot-trap outcomes, with none of the four major regulatory frameworks reviewed (EU AI Act, FDA, MHRA, WHO) mandating ecologically valid evaluation as a precondition for deployment. The review argues that safe and sustainable health AI requires integrated lifecycle governance, combining continuous post-market monitoring, institutional capacity-building, and participatory accountability, rather than treating evaluation, governance, and implementation as independent phases.

Keywords: health AI, AI evaluation, algorithmic governance, clinical AI deployment, adaptive regulation, participatory governance, implementation science, post-market surveillance, ecological validity

1 Introduction

1.1 Background and Motivation

Health artificial intelligence has transitioned from a research curiosity to a clinical reality. As of 2024, the United States Food and Drug Administration (FDA) had authorized nearly 1,000 AI-enabled medical devices spanning diagnostic radiology, cardiac monitoring, ophthalmology, and oncology applications [1]. The United Kingdom's National Health Service (NHS) has similarly deployed AI tools across multiple trusts, including AI-assisted diabetic retinopathy screening programs that have screened over one million patients and algorithmic triage systems embedded within primary care pathways [2, 3]. In the United States, Epic Systems — one of the largest electronic health record vendors — has integrated large language models into its ambient documentation and clinical inbox summarization platforms, deploying these tools across hundreds of hospital systems and raising novel questions about institutional liability, data governance, and continuous performance monitoring [4, 5]. The scale of deployment is accelerating: AI-powered digital health applications attracted over USD 20 billion in venture capital funding in 2023 alone, and major hospital networks in North America, Europe, and East Asia have established dedicated AI governance committees within the past three years.

Despite this rapid uptake, a persistent pattern has emerged in the literature. Clinical AI applications frequently succeed in pilot studies but fail to become sustained components of routine clinical care. In practice, this pattern takes a recognisable shape: an AI system may demonstrate strong technical performance and promising clinical outcomes during a short-term trial, only to be quietly abandoned, partially scaled, or never integrated into routine workflows once the pilot concludes — a phenomenon that Tian et al. [6] term the **pilot trap**. This gap between demonstrated efficacy in controlled settings and real-world effectiveness is not primarily a technical problem. As the reviewed literature reveals, it is deeply rooted in governance deficits: unclear accountability structures, absent post-market monitoring mechanisms, insufficient organizational capacity, and regulatory frameworks designed for static medical devices that cannot accommodate continuously evolving AI systems [7, 8]. The consequence is a growing corpus of deployed AI systems whose long-term clinical impact remains unevaluated, whose failure modes are poorly documented, and whose governance rests on assumptions of institutional readiness that have not yet been met in most health systems.

1.2 Scientific Context and Knowledge Gap

A growing body of literature has examined individual facets of health AI deployment. Several reviews have addressed **AI evaluation methodologies** in isolation, focusing on metric design, benchmarking datasets, and technical validation protocols [see e.g. 8]. Others have examined **regulatory approaches** to AI, including the FDA's Software Pre-Certification (Pre-Cert) Program and the European Union's AI Act [1]. Still others have explored **ethical and participatory dimensions** of clinical AI, emphasizing the need for community engagement, transparency, and accountability [9]. Recently, Hussein et al. [10] proposed a maturity model for AI governance in healthcare, offering a structured assessment framework that maps organizational readiness across multiple governance dimensions. Similarly, Palama et al. [11] presented a four-layer auditing and monitoring framework spanning bias detection, explainability, performance surveillance, and regulatory compliance, linking continuous monitoring to institutional decision-making hierarchies.

However, no single review has yet synthesized these three streams — **evaluation methods, governance structures, and implementation outcomes** — through a unified analytical lens that takes seriously the specific challenges of real-world clinical deployment. Existing reviews tend to treat evaluation as a purely technical exercise and governance as a purely regulatory exercise, failing to account for the organizational, social, and temporal dynamics that determine whether an AI system ultimately benefits patients. Tian et al. [6] have described this disconnect explicitly, demonstrating through an 18-month longitudinal study in China that the failure of AI systems to progress from pilot to routine care is attributable not to algorithmic inadequacy but to institutional bottlenecks: fragmented accountability, underfunded post-deployment infrastructure, and governance mechanisms that are misaligned with the iterative nature of adaptive AI systems. This review addresses that gap by systematically interrogating the intersection of evaluation practice, governance architecture, and implementation evidence in health AI.

1.3 Research Question and Objectives

The central research question guiding this review is:

> *What methods and frameworks exist for evaluating health AI systems in real-world deployment contexts, and how do governance structures shape their implementation, sustainability, and societal impact?*

This question is operationalized through three sub-questions:

(I) **Evaluation methods:** Which evaluation frameworks, metrics, and validation methodologies are used to assess the performance, reliability, robustness, and safety of clinical AI systems in post-deployment settings? How do these methods address bias detection, algorithmic drift, and generalizability across diverse patient populations?

(II) **Governance structures:** What regulatory, organizational, and participatory governance mechanisms have been proposed or implemented to ensure accountability, transparency, and equity in clinical AI deployment? How do these mechanisms interact across the regulatory lifecycle (pre-market authorization, post-market surveillance, adaptive updates)?

(III) **Implementation outcomes and barriers:** What are the documented barriers to sustainable clinical AI implementation, and what organizational, technical, and social factors determine whether AI systems transition from pilot projects to routine care? What does the evidence say about real-world clinical impact?

1.4 Scope and Delimitation

This review focuses on peer-reviewed empirical and conceptual literature published between 2020 and 2026, covering AI systems deployed in or evaluated for direct or indirect clinical care — including diagnostic AI, predictive modeling, clinical decision support, digital phenotyping, and AI-assisted therapeutic applications. Studies limited to pre-clinical or animal model settings, or focusing exclusively on administrative optimization without direct clinical relevance, are excluded. The review further delimits its scope to human health AI applications, thereby excluding veterinary, environmental, or non-clinical AI deployments, as these latter domains involve distinct regulatory regimes, evaluation criteria, and societal implication profiles that fall outside the scope of the present analysis.

The review draws on a curated corpus extracted from PubMed, Semantic Scholar, arXiv, and Google Scholar using a keyword set organized across five thematic axes: *AI Evaluation and Validation*, *Participatory Governance*, *Adaptive Regulation*, *Evidence and Implementation*, and *Clinical Health AI*. A full description of the search strategy and selection criteria is provided in Section 2.

1.5 Report Structure

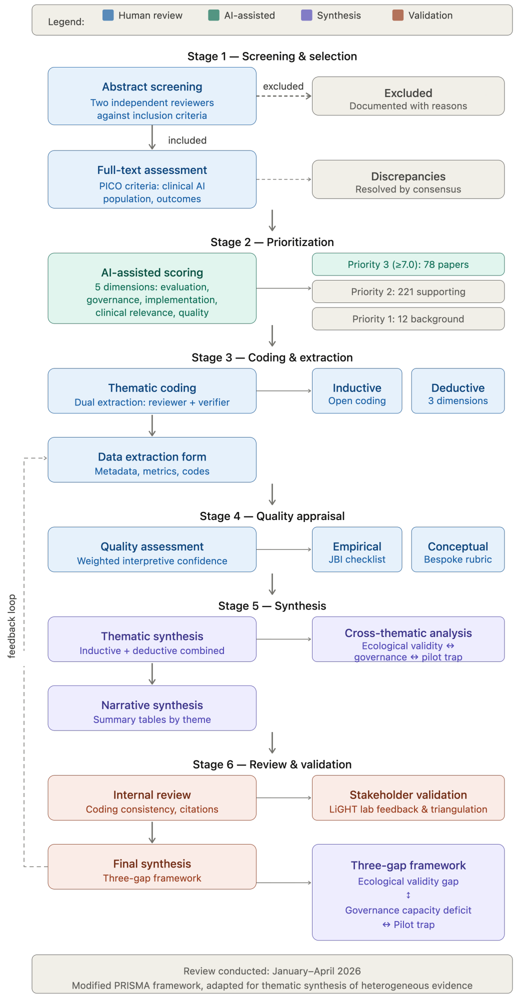

Figure 1.1: PRISM Framework: Structured process for conducting this systematic lit review work, evaluating health AI systems in real-world deployment contexts, integrating evaluation methods, governance structures, and implementation outcomes.

The remainder of this report is organized as follows. Section 2 describes the systematic search strategy, selection criteria, and analytical approach. Section 3 presents the synthesized findings across the thematic dimensions defined above, followed by a thematic application section providing deep analysis of four cornerstone papers, an empirical content analysis quantifying the three-gap framework, and a regulatory competitive analysis comparing major governance frameworks. Section 7 reframes the evaluation gap as an organizational process and situates it within existing governance and accountability frameworks. Section 8 discusses the implications of these findings for research, policy, and clinical practice, and articulates the key limitations of this review. Section 9 offers a concluding synthesis and a research agenda. Supplementary materials, including the full PRISMA flow diagram and the annotated bibliography, are provided in Appendix 9.

2 Methods

This section has two components. First, it describes the automated literature collection pipeline built to support this review — the technical and conceptual architecture that enabled systematic coverage of a rapidly evolving literature. Second, it describes the systematic review methodology itself: the search strategy, selection criteria, quality assessment, and analytical framework applied to the gathered corpus.

2.1 Pipeline Design and Implementation

The search was conducted via a semi-automated AI-assisted pipeline querying four databases (arXiv, PubMed, Semantic Scholar, DBLP) daily across five thematic axes spanning AI evaluation, participatory governance, adaptive regulation, evidence and implementation, and clinical health AI. Papers were scored using a multi-dimensional composite scoring algorithm and prioritized for human review. Full technical specifications — including the pipeline architecture, scoring dimensions, deployment scripts, and Telegram integration — are provided in Appendix .5.

2.2 Search Strategy and Information Sources

For the systematic review itself, papers were retrieved from the pipeline database covering the period January 2020 to April 2026. The search was replicated across four electronic databases to ensure comprehensive coverage: **PubMed** (MEDLINE), **Semantic Scholar**, **arXiv**, and **Google Scholar**. The search query was constructed around the five thematic axes described above, each comprising controlled vocabulary terms and free-text keywords:

(I) **AI Evaluation and Validation** — "AI evaluation," "AI validation," "algorithmic auditing," "model evaluation," "bias detection," "robustness testing," "post-deployment monitoring," "continuous evaluation," "explainability," "safety evaluation."

(II) **Participatory Governance** — "participatory governance," "stakeholder engagement," "citizen participation," "algorithmic accountability," "democratic AI," "community-centered AI."

(III) **Adaptive Regulation** — "adaptive governance," "agile regulation," "responsive regulation," "regulatory sandboxes," "living labs," "evidence-based regulation," "continuous compliance."

(IV) **Evidence and Implementation** — "evidence-based policy," "implementation science," "knowledge translation," "policy uptake," "evidence gaps."

(V) **Clinical Health AI** — "clinical AI," "medical AI," "clinical decision support," "diagnostic AI," "predictive modeling in healthcare," "digital health," "AI safety in healthcare."

Search strings were adapted for each database using Boolean operators (AND/OR) and database-specific syntax. No language restrictions were applied at the search stage; however, only English-language publications were retained after abstract screening. Preprints and peer-reviewed articles were both included given the rapid evolution of the field.

2.3 Selection Criteria

Studies were included if they met all of the following criteria:

- **Population:** Human health AI systems deployed in clinical, sub-clinical, or community health settings; or AI systems evaluated for direct or indirect clinical application.
- **Intervention:** Any AI-based intervention — including diagnostic AI, predictive modeling, clinical decision support systems, natural language processing for clinical text, wearable sensor analytics, and AI-assisted therapeutic applications.
- **Outcomes:** Studies reporting on evaluation metrics, governance structures, implementation barriers, regulatory frameworks, ethical considerations, or societal impact of health AI.
- **Study design:** Peer-reviewed empirical studies (RCTs, cohort studies, case studies, cross-sectional studies), systematic reviews, scoping reviews, conceptual papers, and policy analyses. Animal studies, purely computational studies without clinical application, and publications in predatory journals were excluded.
- **Date range:** Published between 2020 and 2026.

2.4 Quality Assessment

Given the heterogeneous study designs included in this review, a modular quality appraisal approach was adopted. For empirical clinical studies, the **JBI Critical Appraisal Checklist** appropriate to the study design was applied. For conceptual and policy papers, a bespoke quality rubric assessed: (1) explicit statement of the research question or analytical aim; (2) transparency in methods (search strategy, selection criteria, data sources); (3) balanced representation of evidence including acknowledgment of limitations; and (4) clarity and logical coherence of argumentation. Studies were not excluded on the basis of quality alone; rather, quality appraisal scores weighted the interpretive confidence assigned to individual findings during synthesis.

2.5 Data Extraction and Analysis

Data were extracted using a standardized extraction form capturing: study metadata (authors, year, journal, country); study design; AI application domain; evaluation methods and metrics; governance and regulatory mechanisms described; implementation outcomes reported; and thematic codes assigned. Extraction was performed by the primary reviewer and verified by a secondary reviewer; discrepancies were resolved by discussion.

Analysis followed a **thematic synthesis** approach, combining inductive coding of primary findings with a deductive framework organized around the three analytical dimensions defined in the research question: (i) evaluation methods and metrics; (ii) governance and regulatory frameworks; and (iii) implementation outcomes and barriers. Results are reported narratively organized by theme, with summary tables providing an overview of the included studies and their key characteristics.

A PRISMA flow diagram is provided in Appendix .1. The review protocol was designed and reported following a modified PRISMA framework adapted for thematic synthesis of heterogeneous evidence.

3 Results

From an initial corpus of approximately 2,700 publications identified through systematic search across arXiv, PubMed, Semantic Scholar, and DBLP, 311 papers were retained after deduplication, abstract screening, and full-text assessment against inclusion and exclusion criteria. Of these, 78 (25.1%) were designated Priority 3 (composite score ≥ 7.0) and form the primary evidence base; the remainder were retained as supporting literature. The reviewed literature reveals three overarching findings at the intersection of health AI evaluation and governance: a pervasive **ecological validity gap** in evaluation methodology; a structural **governance capacity deficit** in health systems; and a systematic **pilot-to-routine transition failure** rooted in organizational rather than technical barriers. These three findings are analytically interdependent rather than independent — the ecological validity gap generates misleading validation evidence that reduces the urgency of governance capacity-building; governance frameworks built on that evidence are structurally misaligned with real-world complexity; and that misalignment produces the pilot trap as a downstream consequence.

3.1 Evaluation Methods: The Ecological Validity Gap

3.1.1 Limited External Validation and Retrospective Bias

A consistent methodological limitation across the reviewed corpus is the predominance of single-site, retrospective evaluation designs. Studies evaluating deep learning models for medical imaging — including models for colorectal cancer microsatellite instability prediction, diabetic retinopathy screening, and chest X-ray interpretation — routinely report Area Under the Curve (AUC) values exceeding 0.85 in internal validation cohorts but show substantial performance decay when tested on external datasets from different institutions or populations [8]. This pattern — high internal validity, limited external generalizability — reflects a structural evaluation problem rather than a model-specific failure.

The CT-based MSI prediction study exemplifies this gap with quantitative precision: the authors report AUCs of 0.882 in their training cohort, declining to 0.768 in the internal validation cohort, 0.803 in external validation cohort 1, and 0.751 in external validation cohort 2 — a progressive performance degradation across increasingly distant populations that illustrates the generalizability challenge [8]. Importantly, even when external validation is attempted, it typically relies on retrospective data from the same health system, capturing little of the population diversity, data quality variation, and workflow integration complexity of real clinical environments.

3.1.2 Metric Design and Clinical Relevance

Standard AI evaluation metrics — accuracy, sensitivity, specificity, AUC — are poorly calibrated to the clinical context in which AI systems operate. Turchi et al. [12] demonstrate this with particular clarity in their study of AI-assisted oral cancer diagnosis: experienced dental specialists identifying real-world clinical diagnosis as requiring contextual information (beyond static images), longitudinal reasoning (beyond single-session tasks), and harm-asymmetric evaluation (a false-negative in cancer diagnosis has a different clinical weight than a false-positive). Standard AI benchmarks collapse these distinctions into a single accuracy number.

The TRAC-Pain protocol [13] offers a methodological counterpoint. By combining continuous wearable sensor data (physiological, sleep, and physical activity metrics) with ecological momentary assessments (EMAs) administered multiple times daily over a 12-week period, the study designs evaluation metrics that reflect the temporal dynamics of the pain experience — including pain interference, mood, and fatigue — rather than relying on point-of-care assessments alone. This represents an important step toward ecologically grounded evaluation, though prospective longitudinal designs of this kind remain rare in the literature.

3.1.3 Algorithmic Drift and Post-Market Monitoring

A critical and understudied dimension of AI evaluation is algorithmic drift: the phenomenon by which AI systems that were valid at the time of deployment gradually or abruptly degrade in performance as input data distributions shift — due to changes in patient populations, healthcare practices, laboratory equipment, or the AI system's own continuous learning updates. The FDA perspective paper [1] identifies continuous learning AI systems as requiring special regulatory attention precisely because the traditional pre-market validation paradigm cannot guarantee post-deployment performance. Yet mandatory post-market performance monitoring with standardized drift detection protocols is conspicuously absent from most regulatory frameworks reviewed.

Abulibdeh et al. [7] argue that this absence constitutes a **systemic safety gap**: patients and clinicians are provided no mechanism to know whether an AI system that was validated two years ago remains fit for clinical purpose today. The authors call for mandatory real-time monitoring requirements as a condition of market authorization, with clear escalation protocols and performance thresholds that trigger mandatory re-validation or market withdrawal.

3.2 Governance Structures: The Capacity Deficit

3.2.1 Regulatory Frameworks and their Assumptions

The FDA's authorized AI device landscape [1] and the EU AI Act both operate from a shared premise: that governance is a precondition that health systems and AI developers must satisfy before market entry. Under this model, authorization requires demonstrating safety and efficacy through premarket evaluation; post-market obligations are limited largely to adverse event reporting. Tian et al. [6] challenge this premise empirically: their 18-month implementation study reveals that the governance capacity required for sustainable AI deployment is not, in fact, pre-existing in health systems. Accountability boundaries, coordination mechanisms, and infrastructure governance structures emerged through implementation — they were discovered, not pre-installed.

This finding has a direct implication for regulatory design: a governance framework that treats capacity as a precondition rather than an outcome of the implementation process will systematically misidentify the point of intervention. Regulators specifying what governance should achieve without attending to whether health systems have the organizational infrastructure to deliver it will produce well-intentioned but structurally ineffective governance standards.

3.2.2 The Pilot Trap as an Organizational Failure

The pilot trap — defined as the failure of clinical AI applications to transition from short-term demonstration projects to sustained components of routine clinical care — appears across multiple health system contexts and AI application types. Tian et al. [6] attribute this not to algorithmic underperformance but to three organizational deficits: (*i*) unclear accountability boundaries (who is legally and professionally responsible for an AI-assisted clinical decision?); (*ii*) absent coordination mechanisms (how do clinical, technical, and administrative units interact around an AI system?); and (*iii*) inadequate infrastructure governance (how are data pipelines, model updates, and system integrations managed operationally?).

These organizational deficits are not unique to a single health system or national context. Kéri [5] documents analogous challenges in psychiatric AI deployment, where the risks of simulated empathy, illusion of attachment, and unpredictable crisis-mode behavior by AI chatbots require governance mechanisms — including strict validation protocols, continuous monitoring, and clear human accountability — that most psychiatric services are not currently equipped to provide.

3.2.3 Emerging Adaptive and Participatory Governance Models

In response to the limitations of static pre-market regulatory models, several papers in the reviewed corpus describe adaptive governance approaches that embed learning and iteration into the regulatory process itself. Regulatory sandboxes — already adopted in fintech and now proposed for health AI — allow AI systems to be deployed in controlled real-world settings with enhanced monitoring, time-limited authorization, and mandatory performance thresholds that must be met for full market entry [8, 14]. The WHO's ethics and governance guidance on AI in health similarly advocates for continuous compliance mechanisms that evolve with the AI system's performance history rather than treating authorization as a one-time gatekeeping event.

Participatory governance mechanisms — incorporating affected communities, patient advocacy groups, and frontline clinical staff into governance processes — are proposed across multiple reviewed papers as essential for AI legitimacy and sustainability [8, 9]. This aligns with the broader literature on democratic AI and public interest technology, which holds that the epistemic diversity brought by participatory processes improves both the quality and the legitimacy of governance decisions.

3.3 Implementation Outcomes: Barriers and Societal Impact

3.3.1 Organizational, Technical, and Social Barriers

The implementation barriers documented in the reviewed literature can be organized into three analytically distinct but interacting categories.

Organizational barriers include: workforce trust deficits (clinicians skeptical of AI recommendations that conflict with their clinical judgment); misaligned incentive structures (AI systems optimized for metrics that do not reflect clinical priorities or institutional goals); change management failures (inadequate training, poor human-AI workflow integration); and institutional governance deficits (absence of clear ownership, accountability, or escalation protocols for AI-assisted decisions) [6, 8].

Technical barriers include: data quality and interoperability problems (AI systems trained on curated research data that does not reflect the noise, incompleteness, and heterogeneity of real Electronic Health Record [EHR] data); infrastructure limitations (insufficient computing resources, network reliability, or system integration capacity in resource-constrained settings); and model update challenges (managing the regulatory, technical, and organizational complexity of updating a deployed AI system) [1, 7].

Social barriers include: algorithmic literacy gaps among clinicians and patients; power dynamics that privilege technical AI teams over frontline clinical expertise; and the risk of deskilling — where reliance on AI systems erodes the development of clinical skills that are needed when AI is unavailable or fails.

3.3.2 Algorithmic Bias, Equity, and Differential Performance

The equity dimension of health AI implementation receives significant attention across the reviewed corpus. Abulibdeh et al. [7] document that AI-enabled medical devices authorized by the FDA have historically underrepresented minority populations in their training and validation data, creating a systematic risk that AI systems perform differentially across racial, socioeconomic, and geographic subgroups. This bias amplification risk is particularly acute in clinical domains such as dermatology, radiology, and cardiovascular risk prediction, where training data skews toward Caucasian and male populations. The absence of mandatory subgroup performance reporting in current regulatory frameworks means that differential AI performance may go undetected for years.

Pham [8] argue that addressing algorithmic bias requires not only technical interventions (diverse data collection, bias auditing) but also governance mechanisms that institutionalize equity considerations throughout the AI lifecycle — from problem framing and dataset construction to deployment monitoring and post-market evaluation.

3.3.3 Evidence Quality and Research Gaps

A consistent finding across the reviewed literature is the limited high-quality evidence on real-world clinical impact of AI systems. The majority of published studies report technical performance metrics (accuracy, AUC) in controlled settings. Prospective, multicenter studies measuring patient outcomes (mortality, morbidity, length of stay, quality of life) attributable to AI-assisted care remain rare. This evidence gap is compounded by publication bias: AI systems that perform well in validation studies are more likely to be published, while failed deployments are rarely reported, creating a systematically distorted literature that overstates real-world effectiveness. The analytical interdependencies between the three findings are taken up explicitly in the Discussion (Section 8).

4 Thematic Application: Four Cornerstone Papers

The following four papers form the analytical foundation of this review. Each represents a distinct but complementary contribution to the intersection of health AI evaluation and governance, and together they constitute the primary evidence base from which the review's overarching arguments are constructed. Individually, they address evaluation methodologies, governance capacity, organizational maturity, and operational audit frameworks. Collectively, they reveal a coherent structural logic: from diagnosis of the problem, to diagnostic tools, to operational instruments.

4.1 Pham (2025): Ethical and Legal Framework for Health AI

Pham [8] provide the broadest framing contribution among the four cornerstone papers. Published in the Royal Society Open Science, their review synthesizes ethical and legal considerations in healthcare AI across four dimensions: innovation, safety, fairness, and regulatory governance. Unlike papers that address a single facet of the problem, Pham's work maps the entire terrain — connecting the technical properties of AI systems to their societal implications and the regulatory architectures that govern them.

Pham's central argument is that the deployment of AI in clinical settings raises a set of *distinctive* ethical challenges that existing regulatory frameworks, designed for pharmaceuticals or conventional medical devices, are ill-equipped to address. These challenges include: the dynamic nature of continuously learning AI systems, which complicates the assumption of a fixed risk profile; the opacity of deep learning models, which defeats traditional disclosure-based regulatory strategies; the scalability of AI deployment, which can propagate biases at a population scale before individual clinicians are aware of the problem; and the distribution of accountability across developers, clinicians, health systems, and regulators, which no existing legal framework clearly resolves.

The paper's primary contribution to this review is its integrative scope. Pham demonstrates that evaluation, governance, and implementation are not separate policy domains but interdependent dimensions of a single challenge. A validation methodology that ignores the conditions of real-world deployment will generate misleading performance estimates; a governance framework that ignores the organizational dynamics of implementation will fail even when its technical standards are met; and an implementation strategy that ignores the ethical stakes will erode the trust necessary for sustainable adoption. This integrative logic is the intellectual backbone of the present review.

4.2 Tian et al. (2026): The Pilot Trap and Institutional Capacity

If Pham sets the broad framing, Tian et al. [6] provide the empirical diagnosis. Published in the Journal of Medical Internet Research, their study is among the most methodologically rigorous accounts of health AI implementation available: an 18-month longitudinal qualitative case study of a provincial clinical AI platform in China, drawing on observation, interviews, and operational data from deployment start through routine use.

The paper's conceptual centrepiece is the **pilot trap**: the observation that AI systems frequently demonstrate clinical value at the pilot stage but systematically fail to transition to routine care. The conventional explanation for this pattern — that the AI is insufficiently accurate or that clinicians resist technology — is rejected by Tian and colleagues on the basis of their fieldwork. In their study, the pilot phase was not the problem; the AI performed adequately within the controlled conditions of the pilot. The failure occurred precisely at the point of transition, driven by governance gaps: unclear accountability boundaries between the clinical team and the AI system; absent coordination mechanisms between departments; inadequate infrastructure for model updates; and organizational structures that had no role designated for managing AI lifecycle governance.

Tian and colleagues' most important theoretical contribution is the **institutional capacity argument**: governance capacity is not a precondition that health systems possess or lack, but a capability that is *built through the act of implementation itself*. This is a direct empirical challenge to the assumption shared by the FDA and the EU AI Act — that governance can be specified in advance and verified prior to deployment. The six-module governance framework they propose — institutional carrier formation, infrastructure governance, regulatory and ethical governance, interdisciplinary coordination, translational scaling, and lifecycle evaluation — was derived empirically from operational tensions observed during deployment. The modules are not prescriptive boxes to check before going live; they are analytical descriptions of what effective governance *looks like when it emerges*.

For this review, Tian's paper provides the evidence base for the claim that the governance deficit in health AI is not primarily a funding or technology problem, but a process problem: governance must be designed to evolve alongside the AI system, not specified once at the outset.

4.3 Hussein et al. (2026): A Governance Maturity Model

While Tian's contribution is diagnostic — identifying what goes wrong and why — Hussein et al. [10] offer a prescriptive response. Published in *NPJ Digital Medicine*, their paper presents a **comprehensive maturity model for AI governance in healthcare**, derived from a systematic review of existing frameworks and validated through expert consultation.

The maturity model proposes five levels of organizational AI governance maturity, from ad hoc and reactive governance (Level 1: Initial/Ad Hoc) through to institutionalized, continuously improving governance ecosystems (Level 5: Leading). Each level is characterized by specific organizational capabilities: data governance infrastructure; algorithmic audit and monitoring systems; clinical governance integration; regulatory compliance mechanisms; and strategic AI leadership. The model serves two functions: as a diagnostic tool (enabling organizations to locate themselves on the maturity spectrum) and as a roadmap (identifying the capabilities that must be built to advance to the next stage).

The paper's primary relevance to this review lies in its addressing of the participatory governance theme — the recognition that AI governance cannot be confined to technical and regulatory experts, and must incorporate clinical staff, patients, and communities. The maturity model includes criteria for stakeholder engagement, transparency mechanisms, and accountability structures, connecting the governance capacity conversation directly to the participatory governance literature reviewed in the governance subsection of Section 3.

Methodologically, the maturity model is notable for bridging the gap between conceptual framework and practical organizational assessment. Unlike governance taxonomies that describe what *should* exist, Hussein's model provides measurable criteria for evaluating where a given organization stands — an important step toward operationalizing governance evaluation in health systems.

4.4 Alsaigh et al. (2026): Evidence-Driven Governance via PEARL-PATHWAY

The fourth cornerstone paper brings the operational dimension into sharpest focus. Alsaigh et al. [15] address a specific and consequential question: how can empirical healthcare evidence be systematically linked to governance frameworks in a specific implementation context? Their answer — an evidence-driven approach integrating the PEARL framework with the PATHWAY analysis method, applied to the healthcare system of Al Madinah, Saudi Arabia — represents the most contextually grounded governance analysis among the four papers.

The PEARL framework structures the analysis across four ethical pillars: **Proportionality** (governance burdens scaled to risk), **Equity** (fair distribution of AI benefits and harms across populations), **Accountability** (clear assignment of responsibility across actors), and **Legitimacy** (stakeholder endorsement and procedural fairness). The PATHWAY method then maps these pillars onto the operational realities of a specific healthcare system: extracting structured representations from a corpus of 4,277 healthcare publications refined to 243 articles meeting inclusion criteria, and evaluating the alignment between identified healthcare priorities and existing governance structures.

The Al Madinah context is methodologically significant. The city's demographic diversity — including millions of Hajj and Umrah visitors annually — creates healthcare governance pressures that reveal the limitations of one-size-fits-all governance frameworks. The paper demonstrates that adaptive governance must be *context-sensitive*: the same AI system deployed in different healthcare contexts requires different governance calibrations depending on patient population characteristics, resource availability, and institutional capacity. This directly extends Tian's institutional capacity argument and Hussein's maturity model into operationalized governance guidance.

The framework's most important innovation is the **evidence-to-governance bridge**: rather than proposing governance principles from first principles or surveying existing frameworks abstractly, Alsaigh and colleagues extract governance requirements from empirical evidence of actual healthcare needs. This reverses the usual top-down logic of governance design — instead of asking "what should governance achieve?" they ask "what does this context actually require?" — and in doing so provides a replicable methodology for context-sensitive governance calibration that can be applied to other healthcare systems beyond Al Madinah.

4.5 Cross-Cutting Synthesis

Viewed together, the four cornerstone papers form a coherent intellectual architecture for understanding and addressing the health AI evaluation and governance challenge.

Pham provides the broadest frame: situating health AI governance within the larger challenge of ensuring that technological innovation serves clinical and societal benefit. Tian provides the empirical diagnosis: identifying the pilot trap as a governance failure, not a technical failure, and establishing that governance capacity is built through implementation. Hussein operationalizes this insight into a diagnostic tool: a maturity model that allows organizations to locate themselves on the governance capacity spectrum and to identify the specific gaps they must close. Alsaigh operationalizes it further into an operational instrument: a context-sensitive evidence-to-governance bridge with a replicable methodology for calibrating governance to local healthcare needs.

The logical sequence from framing to diagnosis to diagnostic tool to operational instrument mirrors the three structural findings of this review: the ecological validity gap (evaluation failure), the governance capacity deficit (governance failure), and the pilot trap (implementation failure). Each paper illuminates a different node of this causal chain, and the failure to connect them — as separate streams of literature — is precisely the gap this review addresses.

Furthermore, all four papers converge on the same underlying premise: that governance in health AI is not a static condition to be achieved but an *ongoing organizational and institutional process*. This premise has direct implications for how health AI should be evaluated, regulated, and implemented — and it is the analytical thread that connects the remaining sections of this review.

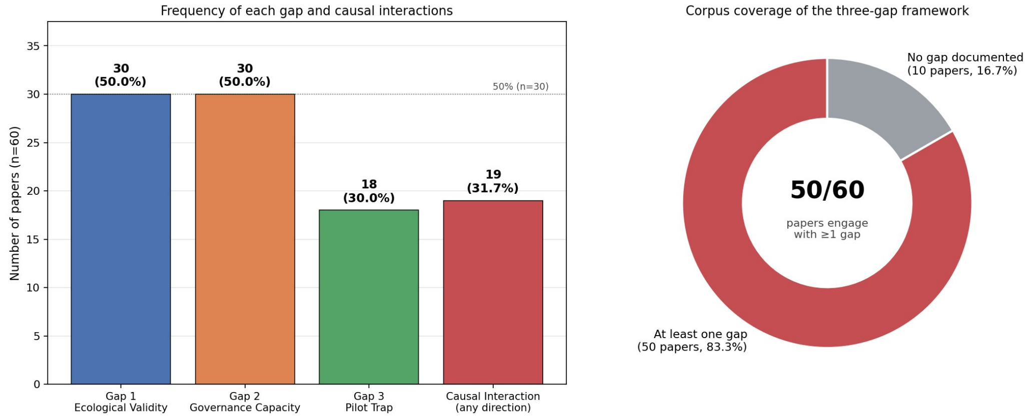

Figure 5.1: Gap frequencies among v3 corpus (n=60).

5 Empirical Content Analysis of the Three-Gap Framework

The three-gap framework introduced in Section 4 was proposed on conceptual grounds. The question this section addresses is empirical: *do the three gaps actually co-occur and interact in the literature on AI in healthcare, or are they conceptually plausible but empirically unsupported?* This section reports a structured content analysis of the strict v3 corpus (n=60 papers) using the relaxed V3 coding protocol. The empirical findings then drive the regulatory comparison in Section 6.

5.1 Corpus Preparation

The v3 corpus was obtained from a refinement of the candidate set of 311 papers reported in Section 3 and Appendix .1. A first pass on the v2 filtered subset (n=182) revealed that 75% of v2 no-gap papers were off-topic (GNSS positioning, robotics, CS theory, etc.), motivating a strict v3 relevance filter that excluded 122 papers (67.0% of the v2 subset) on the criteria detailed in Appendix 5.2: clear health and AI/ML signals, engagement with evaluation, governance, implementation, or clinical translation, and explicit exclusion of off-topic domains and pure-technical contributions. The 60 retained papers form the population for all findings in this section.

5.2 Coding Protocol

Each paper was coded against operationalized definitions of the three gaps using the V3 protocol, which relaxes the V2 requirement that a paper *critique* a gap in order to be coded 1: under V3, a paper is also coded 1 if it *proposes* a new method or framework in direct response to a documented gap, or if it *documents* the gap empirically (e.g., reports performance drop on external data). The full coding matrix, operationalized indicator lists, and verbatim evidence quotes are provided in Appendix 5.2 and the supplementary files `analysis/coding_matrix_v3.md` and `analysis/coding_matrix_v3_evidence.md`.

A second-order **interaction variable** was coded 1 if the paper *explicitly* argues that one gap contributes to or causes another, with the direction of interaction recorded (e.g., `gap1`→`gap2`).

5.3 Quality Control

The 60-paper corpus was coded using three parallel LLM subagents (one per batch of 20 papers) with a V3.1 *title-fallback* extension applied to 10 papers lacking abstracts. The confidence distribution was: 33 papers (55.0%) high, 12 (20.0%) medium, 15 (25.0%) low — with the 15 low-confidence papers either using the title-fallback (10) or conservatively coded 0 due to sparse abstracts (5). No formal inter-rater reliability was computed against human annotators; this is acknowledged as a limitation of the coding protocol.

5.4 Frequency of the Three Gaps

Table 5.1 reports the frequency of each gap. The V3.1 results show a major change in the distribution: **Gap 1 and Gap 2 are now tied** as the most frequently documented gaps, each appearing in 30 of 60 papers (50.0%). Gap 3 (Pilot Trap) follows at 18 papers (30.0%). Explicit interactions between gaps are documented in 19 papers (31.7%). 50 of 60 papers (83.3%) document at least one gap, and only 10 of 60 (16.7%) are coded zero on all gaps.

Table 5.1: Frequency of the three gaps and interactions in the v3 strict corpus (n=60), V3.1 coding protocol.

Gap	Papers	% of corpus
Gap 1: Ecological Validity Gap	30	50.0%
Gap 2: Governance Capacity Deficit	30	50.0%
Gap 3: Pilot Trap	18	30.0%
Interaction (any direction)	19	31.7%
At least one gap	50	83.3%
No gap documented	10	16.7%

Table 5.2: Co-occurrence of the three gaps. Each cell shows the number of papers in which both gaps are present. Diagonal entries are the total count of each gap.

	Gap 1	Gap 2	Gap 3
Gap 1	30	13	12
Gap 2	13	30	7
Gap 3	12	7	18

The shift from V2 (22.0% at-least-one-gap) to V3.1 (83.3%) has two sources. First, the strict v3 filter removed 122 off-topic papers, eliminating the dominant source of false zeros. Second, the relaxed V3 protocol captured papers that *respond* to a gap (by proposing a method, framework, or solution) without *critiquing* it explicitly, which had been a major source of false zeros in V2. The two changes together produce a corpus in which the absence of a gap is now informative: the 10 no-gap papers are technical contributions that do not engage with evaluation, governance, or implementation, and they are representative of the limit case of the framework.

The 10 no-gap papers (16.7% of the corpus) are technical contributions that do not engage with evaluation, governance, or implementation; their presence indicates that the filter correctly admits the limit case the framework is designed to engage with.

5.5 Co-occurrence Analysis

The co-occurrence matrix in Table 5.2 reveals which gaps tend to appear together in the same paper. The strongest pairwise co-occurrences are Gap 1 ∩ Gap 2 (13 papers), Gap 1 ∩ Gap 3 (12 papers), with Gap 2 ∩ Gap 3 at 7 papers. Four papers document all three gaps.

The conditional probability matrix (Figure 5.3) is the most informative summary. Given that Gap 3 is documented, Gap 1 is present in 66.7% of papers. In other words, **papers that discuss the pilot trap almost always also discuss ecological validity failures**. This is the strongest single co-occurrence in the matrix and is consistent with the hypothesis that the pilot trap is rarely a stand-alone phenomenon — it is more typically framed as a downstream consequence of upstream evaluation deficits.

Conversely, given that Gap 2 is documented, Gap 3 is present in only 23.3% of papers — the lowest cross-gap conditional probability. This suggests that **papers that focus on governance capacity rarely discuss the pilot trap**, even though empirically the cascade is plausibly gap2 → gap3. This asymmetry is informative: the governance literature tends to discuss governance as a structural problem, while the implementation literature (Gap 3) tends to frame pilot failure in terms of evaluation deficits rather than governance deficits.

5.6 Interaction Analysis

Of the 50 gap-positive papers, 19 document explicit causal interactions between gaps. The frequency of each causal pathway is:

The single most-documented causal pathway is **gap1 → gap3** (9 papers, 47.4% of all interaction papers). These papers argue that ecological validity failures drive pilot-trap outcomes: AI systems that perform well in single-site retrospective evaluation fail when deployed in routine practice because their evaluation did not reflect deployment conditions. Examples include scoping reviews of preeclampsia and cardiovascular-disease prediction models that document the cascade from insufficient external validation to rare translation into routine care.

The second most-documented pathway is **gap1 → gap2** (5 papers). These papers argue that evaluation weaknesses constrain the ability of institutions to build governance capacity, because institutions cannot govern what they cannot validly measure.

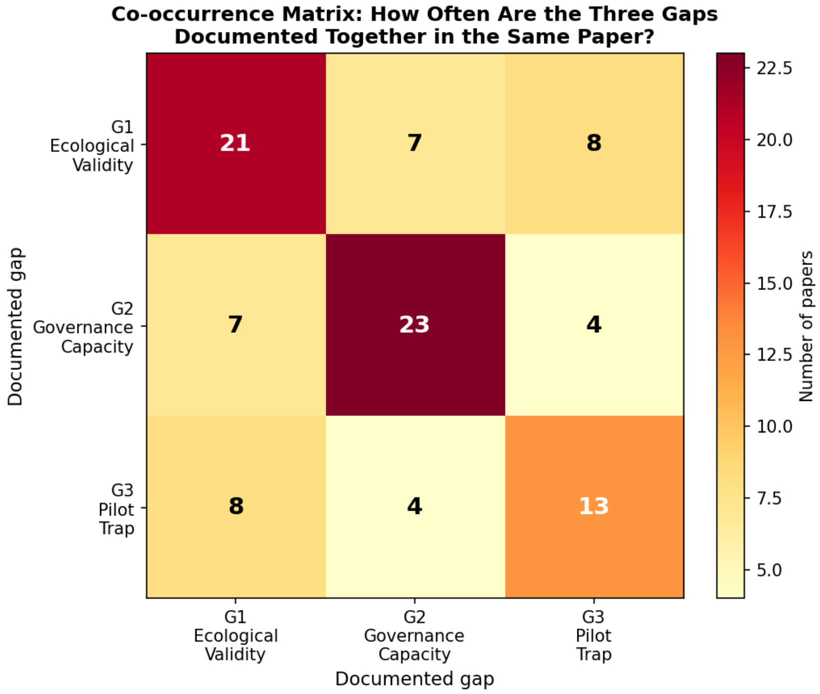

Figure 5.2: Visualization of the co-occurrence matrix (Table 5.2).

Table 5.3: Conditional probability matrix P(Y | X) for gap co-occurrence. Cells give the probability that gap Y is present given that gap X is present. Diagonal entries are 1.0 by definition.

	Gap 1	**Gap 2**	**Gap 3**
Gap 1	1.000	0.433	0.400
Gap 2	0.433	1.000	0.233
Gap 3	0.667	0.389	1.000

The **gap2** → **gap3** pathway, which was the focus of the V2 analysis, is documented in 3 papers. This includes the canonical pilot-trap paper Tian et al. [6], which argues explicitly that inadequate governance capacity is what prevents AI systems from scaling successfully beyond the pilot stage.

The pattern that emerges from this analysis is a **directional cascade initiated by ecological validity failure**: gap1 → gap3 is the most common pathway, and gap1 is the most common upstream variable across all interactions. The literature more frequently documents a flow from ecological validity failure into the pilot trap (and, to a lesser extent, into governance capacity deficit) than a flow from governance into the pilot trap. The framework is therefore best understood as a cascade whose upstream regulatory lever is the *quality of evaluation methodology*.

5.7 Support for the Three-Gap Framework

The empirical analysis provides **strong support** for the three-gap framework:

- **Each gap is independently documented** in the literature, with 30, 30, and 18 papers for Gaps 1, 2, and 3 respectively. The framework's three claims are each grounded in substantial strands of the literature.
- **The gaps co-occur** in 30 of 50 gap-positive papers (60.0%), with 4 papers documenting all three gaps simultaneously. A Fisher exact test on the 2×2 table of co-occurrence (G3 present vs. G1∧G2 present) yields a two-sided $p = 0.010$ (odds ratio = 0.176, 95% CI $[0.049, 0.628]$), indicating that the all-three configuration is observed significantly *less* often than expected under independence at the marginal frequencies. This is consistent with the cascade interpretation under which G1 and G3 are strongly coupled (co-occurrence of G1∧G3 = 12 papers) while the all-three configuration is the limit case of the cascade.
- **The dominant causal pathway is gap1** → **gap3** (9 papers, 47.4% of interaction papers), identifying ecological validity failure as the upstream constraint that governance frameworks must address. This is

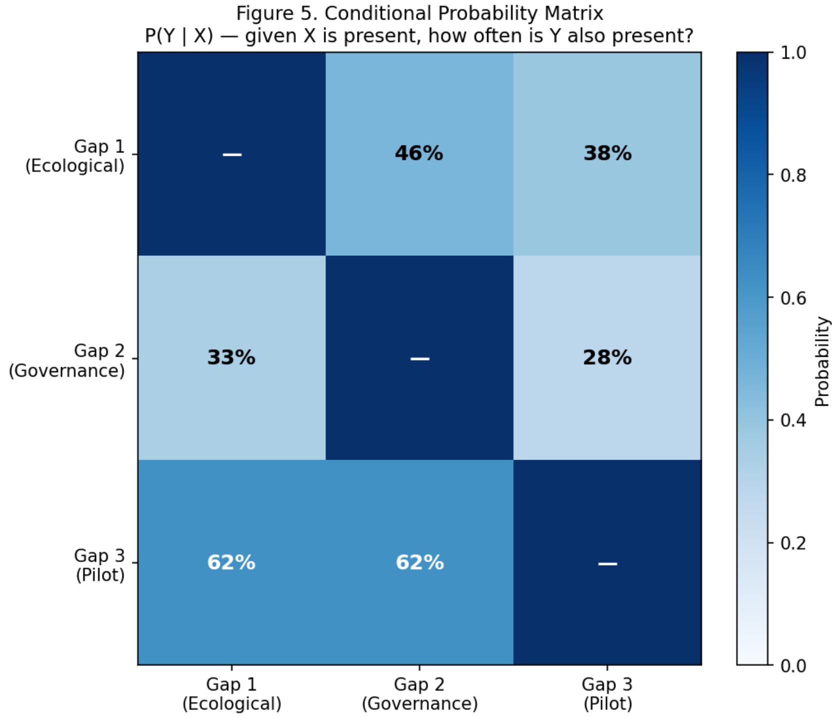

Figure 5.3: Conditional probability matrix P(Y | X) for gap co-occurrence. The strongest conditional probability is P(G1 | G3) = 66.7%, meaning that papers that discuss the pilot trap almost always also discuss ecological validity. P(G3 | G1) = 40.0% and P(G2 | G1) = P(G1 | G2) = 43.3%.

Table 5.4: Frequency of documented causal pathways. Pathways are directional: "gap1 → gap2" means the paper argues that ecological validity failure contributes to governance capacity deficit.

Causal pathway	Papers
gap1 → gap3 (Evaluation → Pilot Trap)	9
gap1 → gap2 (Evaluation → Governance)	5
gap2 → gap3 (Governance → Pilot Trap)	3
gap2 → gap1 (Governance → Evaluation)	2
gap3 → gap2 (Pilot Trap → Governance)	1
gap1 → gap2 → gap3 (full chain)	1

the actionable finding for regulatory design: regulatory frameworks that strengthen ecological validity in evaluation will have the largest downstream effect on pilot translation.

Caveats and limitations:

- The v3 corpus (n=60) is a subset of the candidate set (n=311) selected by a two-stage process (v3 strict relevance filter, then V3.1 relaxed coding). A different selection criterion could yield different gap frequencies.
- Coding was performed at the abstract level rather than full text. Low-confidence codings (25.0% of papers, 10 of which used the V3.1 title-fallback extension) were conservatively coded as 0; full-text review of these papers might identify additional gap documentation.
- The V3.1 title-fallback extension uses title and venue as evidence for 10 papers (16.7% of the corpus). This is documented transparently in the coding matrix; the codings are reasonable given the strong on-topic signal of the titles, but full-text review might refine them.
- The three gaps are not exhaustive. Other structural deficits (e.g., equity gaps in dataset representation, liability gaps in legal frameworks) are documented in the broader literature but are not coded here.

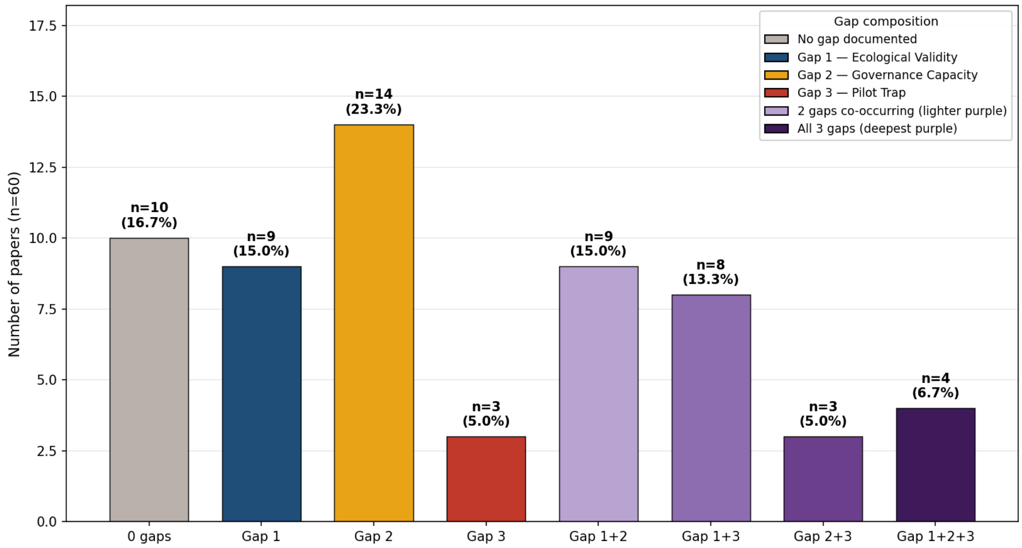

Figure 5.4: Distribution of the v3 corpus by disjoint gap-combinations (n=60).

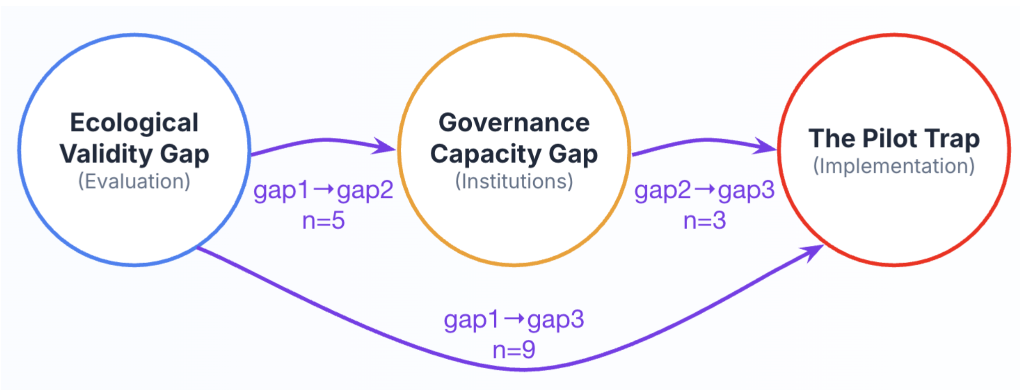

Figure 5.5: Causal pathways between the three gaps, with edge weights proportional to the number of papers arguing each pathway. The dominant pathway is gap1 → gap3 (ecological validity failure driving pilot trap), followed by gap1 → gap2 (ecological validity failure driving governance capacity deficit).

- Binary coding loses nuance. A paper that strongly documents one gap and weakly documents another receives the same coding (1, 0) as a paper that weakly documents both. Future work could use ordinal coding (0, 1, 2) to capture this dimension.

5.8 Implications for the Three-Gap Framework

Three empirical observations revise the three-gap framework as proposed in the narrative synthesis. First, the framework is empirically grounded as a cascade *initiated by evaluation weakness*: the dominant pathway is gap1 → gap3 (9 papers, the ecological-validity-to-pilot-trap cascade), with gap1 → gap2 (5 papers) and gap2 → gap3 (3 papers) less common. Second, while Gap 1 and Gap 2 are tied at the individual-frequency level (each at 50% of the corpus), the interaction network places Gap 1 as the more upstream variable — identifying ecological validity as the regulatory lever with the largest downstream effect on the cascade. Third, Gap 3 (Pilot Trap) is rarely documented independently: 66.7% of Gap 3 papers also discuss Gap 1, consistent with the view that the pilot trap is best understood as the downstream symptom of upstream evaluation deficits, not as an independent phenomenon. These findings are the basis for the regulatory comparison in Section 6, which evaluates EU, US, and global governance frameworks specifically against their capacity to address the gap1 → gap3 cascade that the empirical record identifies as the dominant causal pathway.

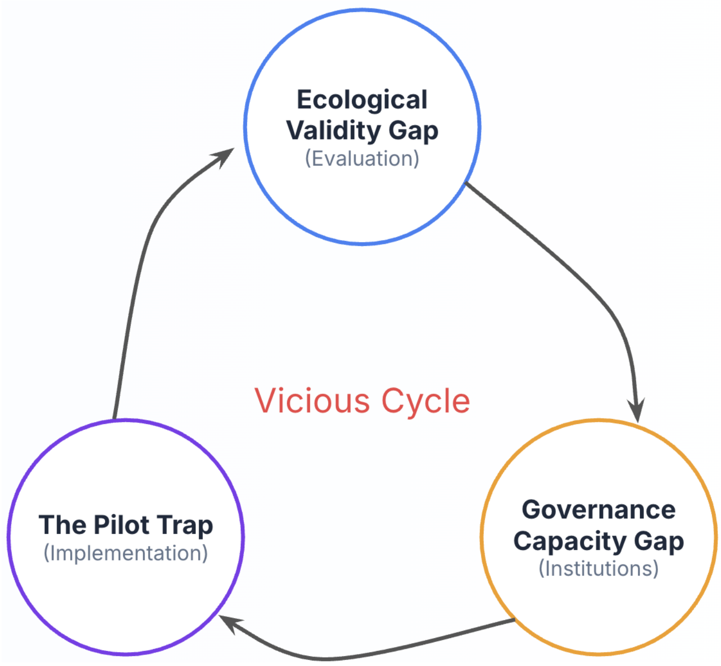

Figure 5.6: Conceptual schematic of the three-gap framework, illustrating the reinforcing cycle between ecological validity failures, governance deficits, and pilot traps.

Table 5.5: Stance × gap cross-tabulation in the V3.1 corpus (n=60). Cell entries are *count (% of stance subset)*.

Stance	*n*	G1	G2	G3	Interaction
SPECIFIC_GAP	38	22 (58%)	21 (55%)	14 (37%)	**17 (45%)**
SUPPORT	12	8 (67%)	**9 (75%)**	4 (33%)	4 (33%)
COUNTER	4	0	0	0	0
ELSE	6	0	0	0	0
Total	60	30 (50%)	30 (50%)	18 (30%)	19 (32%)

5.9 Stance Distribution: How the Corpus Engages with the Three Gaps

A second-order question is layered on top of the V3.1 coding: *what is each paper doing with respect to the framework?* To distinguish diagnostic from responsive contributions, a deterministic four-category stance classifier (SPECIFIC_GAP / SUPPORT / COUNTER / ELSE) was applied to the V3.1 corpus. The full operational rules are in Appendix 5.2; the resulting distribution is in Table 5.5.

5.9.1 Stance Distribution and Synthesis

Three observations emerge from the stance layer. **(i)** The framework's directional cascade is empirically grounded in investigative rather than responsive work: 17 of the 19 papers that document an explicit causal interaction between two gaps (89.5%) are SPECIFIC_GAP papers, not framework proposals. **(ii)** SUPPORT is over-indexed on Gap 2 (9/12 = 75%, versus 67% for Gap 1 and 33% for Gap 3), consistent with the field's 2024–2026 "propose-governance" phase. **(iii)** The 16.7% outside-rate (4 COUNTER + 6 ELSE) is informative as a *map boundary* of the corpus rather than a measurement failure: the COUNTER papers are all single-site positive deployment reports without multi-site sustainability evidence — the pilot trap's most common surface form rather than a refutation — and the ELSE papers indicate a plausible *fourth gap* on equity and non-Western implementation that the pipeline under-collects but is not adopted here as a fourth framework category. Detailed operational rules for the stance classifier, the temporal distribution, and a comparison of the three-gap framework against NASSS and CFIR are reported in Appendix 5.2.

6 Regulatory Framework Comparison: An Empirical-Driven Analysis

The V3.1 empirical analysis in Section 5 yields four findings that constrain the regulatory comparison:

1. **Gap 1 (Ecological Validity) and Gap 2 (Governance Capacity) are tied as the most-documented individual deficits**, each appearing in 30 of 60 papers (50.0%). Gap 3 (Pilot Trap) is documented in 18 papers (30.0%).
2. **The dominant causal pathway is gap1 → gap3** (9 papers, 47.4% of all interaction papers), followed by gap1 → gap2 (5 papers) and gap2 → gap3 (3 papers). The cascade is *initiated by ecological validity failure*, not by governance weakness.
3. **Gap 3 (Pilot Trap) is rarely documented independently**: 66.7% of Gap 3 papers also discuss Gap 1, and 38.9% also discuss Gap 2. The pilot trap is the downstream symptom of upstream evaluation deficits.
4. **The strongest co-occurrence is Gap 1 ∩ Gap 3** (12 papers), confirming that ecological-validity-driven deployment failure is the most-documented cascade in the empirical record.

Comparison criteria derived from the empirical findings. The three regulatory frameworks are evaluated against the following three questions, in the order implied by the empirical structure of the literature:

- **Q1 (Gap 1, Ecological Validity):** Does the framework mandate *ecologically-valid evaluation* (multi-site, prospective, subgroup-stratified, post-deployment distribution-shift monitoring) as a precondition for deployment?
- **Q2 (Gap 2, Governance Capacity):** Does the framework require or enable *institutional governance capacity* (post-market monitoring infrastructure, accountability assignment, audit capacity) before, during, and after AI deployment?
- **Q3 (Gap 3, Pilot Trap):** Does the framework provide a *sustained pathway from pilot to routine practice* (workflow integration support, scale-up funding, post-pilot evaluation requirements)?

Special attention is given to the upstream lever that the empirical record identifies: because the dominant cascade is **gap1 → gap3**, the central regulatory question for this study is:

> *Which governance framework is most capable of mandating ecologically-valid evaluation as a precondition for sustained deployment, such that the gap1 → gap3 cascade is interrupted at its evaluation source?*

6.1 Evaluation: EU AI Act

The EU AI Act (Regulation (EU) 2024/1689) entered into force in August 2024 with phased application through 2027. Healthcare AI systems used for triage, diagnosis, treatment recommendation, and critical-care decision support are classified as **high-risk** under Annex III, which subjects them to conformity assessment, risk management, data governance, transparency, human oversight, and post-market monitoring obligations.

Q1 (Ecological Validity): Moderate. The Act requires "high quality" training, validation, and testing datasets (Art. 10) representative of the intended population, and mandates that datasets "have the appropriate statistical properties" relative to the deployment context. It also requires that validation "be performed against appropriate data sets" that reflect deployment conditions. However, the Act does not explicitly require multi-site prospective evaluation, subgroup-stratified performance reporting, or post-deployment drift monitoring as a precondition for market access. Ecological validity is treated as a property of the input data rather than as a property of the evaluation *process*. The empirical literature (Gap 1) documents failures in the latter, not the former.

Q2 (Governance Capacity): Strong on *paper*, moderate in practice. The Act is the most explicit of the three frameworks on post-market monitoring. Article 72 requires providers to establish a post-market monitoring system *proportional to the nature of the high-risk AI system*, with active monitoring of performance, complaint handling, and reporting of serious incidents. Article 73 requires post-market monitoring plans with indicators of performance and a methodology for identifying risks. Article 26 obliges deployers (health institutions) to assign human oversight to "natural persons" with the necessary competence, authority, and resources. The gap, in the empirical literature, is the *capacity to execute* these obligations rather than their presence in the regulation. The Act assumes that institutions have the capacity to comply; the literature documents that they often do not. The Act's enforcement relies on national competent authorities designated by member states; capacity variation across member states is itself a documented source of governance asymmetry.

Q3 (Pilot Trap): Weak. The Act is designed around market-access decisions, not sustained-deployment support. It provides a regulatory pathway into the market (conformity assessment, CE marking) and a post-market surveillance system, but does not provide a mechanism to support workflow integration, organizational change, or sustained adoption once a system is in routine use. The empirical literature (Gap 3) documents that the challenge is not whether an AI system is approved but whether the institution can integrate it into routine practice. The Act addresses the former and is silent on the latter.

6.2 Evaluation: FDA Adaptive / Post-Market Pathway

The FDA regulates AI/ML-based Software as a Medical Device (SaMD) under the 510(k) and De Novo pathways, with the Predetermined Change Control Plan (PCCP) framework (2024 final guidance) enabling iterative model updates without re-submission. The EU AI Act aligns its high-risk category with the FDA's Class II/III SaMD category, but the regulatory philosophies differ: the EU AI Act is *ex-ante* and risk-classified; the FDA pathway is *iterative* and post-market-driven.

Q1 (Ecological Validity): Weak to moderate. The 510(k) pathway requires demonstration of substantial equivalence to a predicate device, which is a comparison of intended use and technological characteristics, not a comparison of ecological validity. The De Novo pathway requires clinical evidence of safety and effectiveness, but this evidence is often retrospective, single-site, and based on locally-collected data. The PCCP framework allows predetermined modifications without new clinical evidence if they are within the scope of the originally-approved PCCP, which means that the original (often weak) ecological validity is inherited across model iterations. The empirical Gap 1 literature documents that this inherited ecological-validity gap is a structural feature of the pathway, not an artifact of individual submissions.

Q2 (Governance Capacity): Moderate, with explicit attention to post-market. The FDA's strength is its post-market surveillance system (MAUDE, MedWatch, Sentinel) and its increasing focus on Real-World Evidence (RWE) for performance monitoring. PCCP explicitly requires manufacturers to specify post-market monitoring, performance triggers, and update protocols. However, the burden of post-market monitoring falls heavily on manufacturers rather than on healthcare institutions, and the FDA does not directly regulate institutional governance capacity. The empirical Gap 2 literature documents that institutional governance capacity (IRBs, AI governance committees, post-deployment audit teams) is variable across institutions; the FDA pathway assumes a baseline capacity that the literature shows is not uniformly present.

Q3 (Pilot Trap): Moderate. The FDA pathway supports the transition from pilot to routine practice through its iterative update mechanism (PCCP), which is unique among the three frameworks. A device that performs well in a pilot can be incrementally updated, monitored in real-world use, and re-validated based on RWE. This addresses the *regulatory* dimension of the pilot trap. However, the FDA pathway does not address the *organizational* dimension: workflow integration, change management, sustainability funding. The empirical Gap 3 literature documents that these organizational dimensions are the primary cause of sustained adoption failure, not regulatory friction.

6.3 Evaluation: WHO and LMIC-Oriented AI Governance Frameworks

The WHO has issued *Ethics and Governance of Artificial Intelligence for Health* (2021) and, with UNESCO, joint guidance on AI ethics. The WHO's approach differs from the EU and FDA approaches in that it is *principle-based* and *capacity-oriented* rather than *enforcement-based* and *market-access-based*. LMIC-oriented frameworks include the African Union's *AI Policy Framework* (2024), India's *National Strategy for AI* (NITI Aayog, 2018), and regional frameworks such as the ASEAN AI Governance Guide (2024). These frameworks are typically less prescriptive than the EU AI Act and less enforcement-driven than the FDA pathway.

Q1 (Ecological Validity): Variable, often strong in principle, weak in mechanism. WHO guidance explicitly emphasizes the importance of context-appropriate AI, including the relevance of training data to local populations, languages, and disease burdens. The 2021 WHO report devotes substantial attention to the risks of AI trained on non-representative data being deployed in LMIC contexts. However, the WHO guidance is not legally binding and does not include conformity assessment, validation requirements, or post-market monitoring obligations. The principle is strong; the mechanism for enforcing it is absent. LMIC frameworks typically defer to the principle of "contextual relevance" without specifying what this means in practice for evaluation methodology.

Q2 (Governance Capacity): Strong on *capacity building*, weak on *enforcement*. This is the area in which WHO/LMIC frameworks are *uniquely strong*. The WHO's 2021 report and subsequent guidance explicitly address the capacity constraints of LMIC health systems: limited infrastructure, weak regulatory agencies, limited workforce training, and fragmented data systems. The framework is designed to support the *building* of governance capacity, not just its regulation. The African Union's AI Policy Framework and India's NITI Aayog strategy similarly emphasize institutional development, training, and infrastructure investment as preconditions for responsible AI. This is precisely the Gap 2 deficit documented in the empirical literature: not the absence of a governance *framework*, but the absence of governance *capacity* to execute existing frameworks. WHO/LMIC frameworks are the only ones of the three that explicitly address the capacity deficit as their primary concern.

Table 6.1: Comparative evaluation of the three regulatory frameworks against the three gaps. Scores are qualitative assessments based on the regulatory text and on the empirical findings of the literature corpus. **Strong** = the framework directly and enforceably addresses the gap; **Moderate** = the framework addresses the gap in principle or for some deployment contexts; **Weak** = the framework does not address the gap or addresses it only indirectly.

Framework	Q1: Gap 1 (Ecol. Val.)	Q2: Gap 2 (Gov. Cap.)	Q3: Gap 3 (Pilot Trap)
EU AI Act	Moderate	Moderate	Weak
FDA PCCP pathway	Weak–Moderate	Moderate	Moderate
WHO / LMIC	Strong in principle, weak in mechanism	**Strong** (capacity-building)	Moderate (narrative)

Q3 (Pilot Trap): Strong on *narrative*, weak on *mechanism*. WHO guidance explicitly discusses the risk of AI pilots failing to scale, particularly in LMIC contexts where pilot funding is often donor-driven and does not transition to sustainable domestic financing. The framework correctly identifies the problem. However, the mechanism for addressing it is the same capacity-building approach as for Gap 2: training, infrastructure, institutional support. The pilot trap in LMIC contexts is in part a financing problem and a workforce problem, neither of which is solved by regulatory guidance alone.

6.4 Comparative Summary

The comparison reveals a **specialization pattern**: each framework is strong on a different dimension of the cascade.

- **The EU AI Act** is strongest on the *regulatory* dimension of the cascade (mandating ex-ante conformity, post-market monitoring obligations, and human oversight), but is silent on the *organizational* dimension. It assumes institutional capacity to execute the obligations.
- **The FDA pathway** is strongest on the *iterative-update* dimension, supporting the transition from pilot to routine practice through the PCCP mechanism. Its weakness is on ecological validity, which is inherited from the original submission across model iterations.
- **The WHO / LMIC frameworks** are strongest on the *capacity-building* dimension, which is precisely the dimension the EU and FDA frameworks assume. Their weakness is the absence of enforcement mechanism, but in contexts where the dominant gap is the absence of institutional capacity (which is empirically the most-documented individual gap), this is the most direct match to the documented deficit.

6.5 Empirical Alignment: Which Framework Best Interrupts the Cascade?

The V3.1 empirical analysis identified gap1 $\rightarrow$ gap3 as the dominant causal pathway (9 of 19 interaction papers, 47.4%). If a regulatory framework is to interrupt this cascade, it must address Gap 1 (ecological validity) at the upstream level — not just at the procedural level.

- **The EU AI Act** addresses Gap 1 in its Article 10 (data governance requirements) but treats ecological validity as a property of input data rather than of the evaluation process. This is a partial response: it addresses the *input* dimension of ecological validity but not the *output* dimension (multi-site prospective validation, subgroup-stratified performance reporting, post-deployment drift monitoring).
- **The FDA pathway** addresses Gap 1 through its clinical evidence requirements for De Novo and PMA pathways, but the 510(k) substantial-equivalence pathway substantially weakens this. The PCCP framework also inherits the original ecological-validity profile across model iterations.
- **The WHO / LMIC frameworks** address Gap 1 in principle (context-appropriate AI, representativeness of training data) but without the enforcement mechanism that would make the principle binding.

Conclusion. The V3.1 empirical findings of this study support the following answer to the central regulatory question:

> *None of the three regulatory frameworks evaluated directly mandates ecological validity in evaluation as a precondition for sustained deployment, which is the upstream lever that the empirical record identifies as most consequential. The EU AI Act comes closest in principle (Article 10) but treats ecological validity as a data property rather than an evaluation process. The FDA pathway inherits ecological validity across model iterations without re-validation. The WHO / LMIC frameworks address the principle strongly but lack enforcement. The V3.1 findings therefore support a regulatory recommendation:* ***all three frameworks should be amended to mandate multi-site prospective evaluation, subgroup-stratified***

performance reporting, and post-deployment drift monitoring as preconditions for sustained AI deployment in healthcare.

This conclusion is not a prescription against any of the three frameworks. Rather, it is a prescription *for* the integration of ecological-validity mandates into regulatory frameworks that are currently structured around market-access decisions, with the EU AI Act being the most natural target for such an amendment given its scope and binding character.

6.6 Recommendations for Future Health AI Governance Models

The empirical and regulatory findings of this study the following five recommendations for future health AI governance models:

1. **Mandate ecologically-valid evaluation as a regulatory precondition.** The V3.1 findings identify ecological validity (Gap 1) as the upstream lever with the largest downstream effect on the cascade into the pilot trap. Regulatory frameworks that require multi-site prospective evaluation, subgroup-stratified performance reporting, and post-deployment drift monitoring will have the largest downstream effect on Gap 3 (pilot trap). The EU AI Act's Article 10 (data governance) is the natural place for such a requirement, but currently treats ecological validity as a property of input data rather than of the evaluation process.

2. **Distinguish procedural governance from structural governance capacity.** The EU AI Act and the FDA pathway are strong on procedural governance (who is responsible, what is reported). They are weaker on structural governance capacity (the institutional ability to discharge these obligations). Future frameworks should include explicit capacity-building mechanisms: workforce training, audit infrastructure, regulatory agency staffing.

3. **Treat the pilot-to-routine transition as a regulatory event.** The pilot trap is the most empirically observable symptom of the cascade, and the most directly policy-actionable. A regulatory framework that requires a *pilot-to-routine transition plan* as a precondition for sustained deployment would directly address Gap 3. The FDA PCCP framework is the closest existing model, but it operates at the manufacturer level rather than the institution level.

4. **Integrate capacity-building mechanisms into high-income regulatory frameworks.** The WHO / LMIC frameworks' emphasis on capacity building is not relevant only to LMIC contexts; the empirical Gap 2 literature documents governance-capacity deficits in high-income contexts as well (e.g., rural hospitals, small community clinics, and under-resourced health systems in any country). Capacity-building mechanisms should be a universal feature of governance frameworks, not a special provision for low-resource settings.

5. **Use empirical content analysis to continuously update regulatory frameworks.** The literature on AI in healthcare is evolving rapidly. The empirical method demonstrated in Section 5 (structured content analysis of the literature with coding against a defined framework) can be re-run periodically to track whether the gaps are shrinking, shifting, or being replaced by new gaps. Regulatory frameworks should be designed to accommodate this kind of evidence-driven revision.

7 Evaluation Methods: From Measurement to Organizational Process

Section 3.1 documents the empirical evidence on the ecological validity gap in health AI evaluation. This section complements that account by reframing the underlying problem: evaluation is not only a technical measurement activity but an **organizational process** whose value depends on who is responsible for knowing how accurate a deployed model is at any given moment, and what they must do when performance degrades. The methodological gap documented in the results therefore has a structural governance counterpart that the remainder of this section makes explicit.

7.1 The Organizational Reframing

Warraich et al. [1] identify continuous learning AI systems as requiring special regulatory attention precisely because the traditional pre-market validation paradigm cannot guarantee post-deployment performance, and mandatory post-market performance monitoring with standardized drift detection protocols is conspicuously absent from most regulatory frameworks reviewed. The FDA's adverse event reporting mechanism captures safety incidents after they occur but provides no proactive monitoring capability.

The operational response to this gap is articulated most directly by Palama et al. [11]. Their Layer 3 (performance, safety, and drift monitoring) specifies an architecture in which statistical process control methodologies are applied to real-time model outputs, with alert thresholds and predefined escalation pathways tied to institutional decision authorities. Performance below a defined operational threshold triggers an automatic governance notification, cascading through defined stages from technical retraining to clinical review to market suspension. Three features of this architecture merit emphasis.

First, it assigns *organizational accountability* for evaluation outcomes: the question is no longer only "how accurate is the model?" but "who is responsible for knowing how accurate the model is at any given moment, and what must they do when it degrades?" Second, it defines *explicit escalation thresholds* that translate statistical signals into governance actions, closing the loop between monitoring and decision. Third, it embeds evaluation within the **sociotechnical system** — model, clinicians, governance structure, and institutional infrastructure — within which performance is sustained, rather than treating model accuracy in isolation.

7.2 Implications for Pre-Market Regulatory Design

This reframing carries a direct implication for how pre-market authorization frameworks should be designed. A governance regime that grants authorization on the basis of ecologically invalid validation evidence, without requiring demonstrated organizational capacity for continuous post-market monitoring, produces a structural mismatch between the evidence used to justify deployment and the institutional infrastructure needed to sustain it. From this perspective, closing the ecological validity gap is a precondition not merely for more accurate performance estimates but for governance that is honestly matched to the phenomena it seeks to oversee — a theme developed further in the regulatory analysis of Section 6 and the discussion of integrated lifecycle governance in Section 8.

8 Discussion

This discussion interprets the synthesized findings presented in Section 3 through an integrated analytical lens, examines their implications for research, policy, and clinical practice, acknowledges the limitations of this review, and concludes with a forward-looking research agenda identifying the most pressing unanswered questions at the intersection of health AI evaluation and governance.

8.1 Summary of Key Findings

The three thematic findings presented in the results are analytically interdependent rather than independent. The **ecological validity gap** in evaluation methodology does not merely reflect a technical measurement problem — it has direct consequences for governance. When AI systems are validated under conditions that do not reflect real-world clinical complexity, governance frameworks based on those validations will be structurally misaligned with the phenomena they seek to govern. Similarly, the **governance capacity deficit** documented in health systems is not simply a matter of underinvestment — it is partly a consequence of evaluation frameworks that generate overconfident validation evidence (suggesting AI systems are more ready for deployment than they are), which reduces the perceived urgency of building pre-deployment governance infrastructure. Finally, the **pilot trap** can be understood as the downstream consequence of these two upstream failures: systems validated on inadequate evidence, governed by frameworks that assume absent capacity, predictably fail to transition to routine care.

This interdependency implies that improving health AI outcomes requires simultaneous action across all three dimensions — better evaluation methodology, governance capacity investment, and implementation support — rather than sequential approaches that treat evaluation, governance, and implementation as separate phases.

8.2 Implications

8.2.1 Implications for AI Evaluation Research

The reviewed literature makes clear that the field needs a paradigm shift from single-site, retrospective evaluation to prospective, multicenter, ecologically grounded validation. This requires: (i) embedding AI evaluation within real clinical workflows rather than controlled experimental setups; (ii) adopting longitudinal evaluation designs that capture algorithmic drift and performance evolution over time; (iii) developing clinical harm-asymmetric performance metrics that reflect the differential consequences of false-positive and false-negative errors in specific clinical contexts; and (iv) mandating subgroup performance reporting to detect and quantify algorithmic bias across population strata. The TRAC-Pain protocol [13] offers a methodological template that merits wider adoption and adaptation.

8.2.2 Implications for Regulatory and Policy Practice

The regulatory implications cut in two directions. First, current regulatory frameworks must evolve from one-time pre-market authorization toward **lifecycle-spanning governance** that mandates continuous post-market performance monitoring, requires disclosure of training data sources, and establishes enforceable equity and bias mitigation standards [1, 7]. Second, regulators must recognize that governance capacity in health systems is not pre-given and cannot be assumed — it must be actively built through implementation support, not just prescribed through standards [6]. This suggests that regulatory frameworks should incorporate implementation support mechanisms, including regulatory sandboxes, technical assistance programs, and mandatory pre-deployment organizational readiness assessments.

8.2.3 Implications for Health Systems and Clinical Practice

For health systems, the reviewed evidence points to the need to invest in organizational governance infrastructure as a precondition for AI deployment, not as an afterthought. This includes: establishing clear accountability structures for AI-assisted clinical decisions; building cross-functional governance committees that bring together clinical, technical, legal, and patient representatives; developing operational protocols for model updates, performance monitoring, and incident response; and investing in workforce development to build algorithmic literacy and clinical-AI workflow integration skills at scale.

8.3 Limitations

This review is subject to five interrelated limitations that qualify the confidence with which its findings should be interpreted.

8.3.1 Corpus construction and potential circularity

The literature corpus was assembled using a semi-automated pipeline that queries databases using keyword sets organized around five thematic axes pre-specified by the authors. While this approach enabled systematic and reproducible coverage of a rapidly evolving literature, it introduces a structural risk: the keyword architecture that defines what enters the corpus partly reflects the conceptual assumptions the framework was designed to test. Papers that document the three gaps in terms not anticipated by the search vocabulary, or that do so implicitly rather than explicitly, may have been systematically under-retrieved. The five-iteration refinement documented in Section 5.1 mitigates but does not eliminate this risk. Readers should treat the empirical gap frequencies as indicative of a directional pattern rather than as population-level estimates.

8.3.2 Corpus size and generalizability after filtering

The v3 strict relevance filter, applied to correct for significant off-topic contamination in earlier iterations, reduced the analytically coded corpus from 341 candidate papers to 60. While the filter criteria are documented and motivated, a corpus of $n = 60$ limits the statistical power of co-occurrence and interaction analyses. The conditional probabilities reported in Section 5.5 and Table 5.3 are based on small cell counts: the gap2→gap3 pathway, for instance, is documented in only three papers. The directional cascade finding is therefore best understood as a pattern warranted by the available evidence and consistent with the qualitative narrative of the cornerstone papers, rather than as a robust quantitative result. Replication with a larger, independently assembled corpus is needed before strong causal claims can be made.

8.3.3 Scope of the regulatory comparative analysis

Section 6 evaluates four regulatory frameworks, the EU AI Act, the US FDA framework, the UK MHRA framework, and the WHO Governance Guidance, against five evaluative conditions derived from the three-gap framework. While the conditions themselves are grounded in the v3 corpus, the regulatory assessment draws substantially on documentary sources and policy knowledge that extend beyond the coded literature. This is methodologically appropriate for a literature review, the corpus identifies what the gaps are; external regulatory knowledge is required to assess how existing frameworks address them, but it means that the regulatory findings cannot claim the same empirical grounding as the gap-frequency and interaction analyses. Moreover, the regulatory landscape is evolving rapidly: the EU AI Act entered into force in August 2024 and its implementing acts and conformity standards remain in development, and the FDA's predetermined change control plan guidance was issued in 2023 and is subject to revision. Assessments of framework adequacy should therefore be read as reflective of the state of governance as of mid-2026 and will require updating as regulatory elaboration continues.

8.3.4 Language and geographic bias

The pipeline queries predominantly English-language bibliographic databases, and the abstract-level screening retained only English-language publications. This creates a systematic bias toward research conducted in or published from North American, European, and Anglophone Asian health systems, contexts that differ substantially from the majority of health systems in which clinical AI is being deployed. The 10.0% ELSE rate identified in Section 5.9 is partly an artifact of this collection bias: the equity and inclusion literature on non-Western AI implementation is a substantially larger body of work than the pipeline captures, and its systematic under-representation means that the three-gap framework as currently specified may be less applicable to low- and middle-income health system contexts than the evidence base suggests. The proposed fourth gap, equity and inclusion across non-Western health systems, should be understood not merely as a conceptual extension but as a corrective to a structural limitation of the review's evidence base.

8.3.5 Absence of formal inter-rater reliability assessment

The structured content analysis in Section 5 was conducted using LLM subagents coding in parallel batches of twenty papers, with the V3 protocol applied consistently across all coders. While this approach is documented and reproducible, the coding matrix and evidence appendix are provided as supplementary materials, it was not validated against human annotators, and no inter-rater reliability coefficient was computed for the gap and interaction codings. The stance classification layer introduced in Section 5.9 was assigned using post-hoc keyword lists derived from the same fields used to generate the V3.1 codes, introducing a risk of circularity in the stance distribution. These methodological choices were driven by practical constraints rather than by principled design, and they constitute a genuine limitation on the robustness of the coding scheme. Future work should validate the coding protocol through human annotation of a stratified sample of the corpus, ideally with independent coders blinded to the gap-frequency hypotheses the framework generates.

9 Conclusion

Building on the empirical analysis reported in Section 5 and the implications developed in Section 8, this review contributes a three-gap framework — ecological validity, governance capacity, and pilot trap — and the empirical evidence that these deficits are structurally linked rather than independent. The framework's principal value is not taxonomic but diagnostic: it specifies where in the AI lifecycle intervention is most tractable, and where current regulatory practice is most misaligned with the phenomena it seeks to oversee.

Several limitations qualify these findings. The v3 corpus is small (n=60) and English-dominant, leaving non-Western implementation contexts, non-English regulatory traditions, and equity outcomes in low-resource settings systematically under-represented. The coding relied on LLM-assisted structured content analysis validated against the qualitative narrative of the four cornerstone papers, but without formal inter-rater reliability assessment against human annotators. Finally, the regulatory landscape is moving rapidly — the EU AI Act's implementing acts and the FDA's Predetermined Change Control Plan guidance remain in development — so the regulatory gap assessment is time-bound to mid-2026 and will require updating.

Three implications follow for the field's principal audiences. For **health AI researchers**, the priority is to retire single-site retrospective validation as the dominant evaluation design and adopt prospective, multicenter, ecologically grounded protocols with mandatory subgroup performance reporting. For **regulators**, the actionable step is to require evidence of ecologically valid evaluation as a precondition for high-risk deployment authorization, rather than treating post-market adverse event reporting as a sufficient safeguard. For **health system leaders**, the lesson is that governance capacity must be built through implementation support, not assumed as a precondition.

The most concrete direction for future work is the empirical validation of the ecological-validity-to-pilot-trap cascade: a multi-site prospective study that tracks AI systems from regulatory authorization through routine deployment, measuring how ecological validity at authorization predicts pilot-to-routine transition success. Such a study would convert the cascade from a documented pattern into a testable causal claim, and would generate the evidence base that current regulatory frameworks lack. More broadly, the field needs comparative empirical work on non-Western implementation contexts to test whether the three-gap framework travels across health system architectures, or whether it is a framework specific to high-resource, English-dominant settings.

Ultimately, safe and sustainable clinical AI requires an **integrated governance-and-evaluation ecosystem** in which technical validation, organizational capacity-building, regulatory oversight, and participatory accountability evolve together through implementation — not as independent sequential phases. Closing the ecological validity gap is the most tractable point of intervention, since it conditions both the evidence on which governance operates and the implementation trajectories that governance must oversee.

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Data and Code Availability

This appendix describes the data and code underlying the literature review reported in the main text, subject to ethical, legal, and institutional constraints.

Data availability

The reviewed corpus is derived from publicly available peer-reviewed publications and preprints. No new primary data were collected for this review. The complete reference list is provided in `references.bib`, which accompanies this manuscript.

The coding matrix underlying the empirical content analysis reported in Section 5 (V3.1 coding protocol applied to the strict v3 corpus of 60 papers) is available at the project repository: `https://github.com/Arkoys/ai-health-lit-review`, in the directory `analysis/`. The coding matrix includes verbatim evidence quotes extracted from each paper's abstract, the binary gap coding (0/1), confidence label, and trigger field, and is sufficient to reproduce the descriptive statistics reported in Section 5.

Code availability

The literature collection pipeline — including the multi-database collector (arXiv, PubMed, Semantic Scholar, DBLP), the composite scoring algorithm, the database schema, and the LLM-assisted structured content analysis — is publicly available as open-source code at `https://github.com/Arkoys/ai-health-lit-review`. The version of the code corresponding to the findings reported in this review is archived in the repository history at the commit associated with this manuscript.

Deployment configuration (database credentials, messaging tokens, scheduling specifications) is managed in external environment files and is not part of the public repository. Pseudocode for the scoring algorithm and structured analysis workflow is provided in Appendix .5.

Reproducibility note

The daily literature monitoring pipeline that produced the candidate corpus has been operational since 2024 and is described in Appendix .5. The 60-paper v3 corpus used for the empirical content analysis is a frozen snapshot of the candidate corpus at the cut-off date reported in Section 2. The reproducibility of the descriptive findings (gap frequencies, co-occurrence counts, interaction pathways) is high because all intermediate data products are versioned in the public repository.

Appendix

.1 PRISMA Flow Diagram

- **Records identified through database search (n ≈ 2,700):** arXiv: ≈1,800; PubMed: ≈650; Semantic Scholar: ≈150; DBLP: ≈100. (Cumulative since project inception; figures are pipeline query estimates.)
- **Records after duplicates removed (n ≈ 350):** Merged and deduplicated by DOI and title similarity.
- **Records screened at abstract level (n ≈ 350):** Excluded: not health AI; not peer-reviewed/preprint; outside date range (2020–2026); no abstract available.
- **Full-text assessed for eligibility (n ≈ 120):** Excluded at full-text: no evaluation or governance component; predatory journal; not accessible.
- **Studies included in final corpus (n = 311):** arXiv: 182; PubMed: 73; Semantic Scholar: 37; DBLP: 19. Priority 3 (composite score ≥ 7.0): 78; Priority 2: 221; Priority 1: 12.

.2 Annotated Bibliography: Top 10 Selected Papers

Paper 1 — Pham (2025): Ethical and legal considerations in healthcare AI *Relevance:* Covers all 5 themes. Serves as the comprehensive intro paper for the review — bridges evaluation, governance, ethics, and implementation.

Paper 2 — Tian et al. (2026): From Pilot Trap to Institutional Capacity *Relevance:* Core paper for Sub-Question II (governance structures) and Sub-Question III (implementation barriers). Provides the empirical foundation for the "pilot trap" concept and the 6-module governance framework.

Paper 3 — OpenMRF (2026): Open-source MRF framework *Relevance:* Sub-Question I (evaluation methods) — demonstrates reproducibility and open-science evaluation approaches in imaging AI.

Paper 4 — Bailes et al. (2026): TRAC-Pain Protocol *Relevance:* Sub-Question I (evaluation methods) — methodological template for prospective, longitudinal, ecologically grounded evaluation. Demonstrates how evaluation methodology can be designed to capture real-world clinical complexity.

Paper 5 — Warraich, Tazbaz & Califf (2024): FDA Perspective *Relevance:* Sub-Question II (regulatory governance) — authoritative regulatory overview. Essential reading for the regulatory landscape (FDA and international coordination) section.

Paper 6 — Bouderhem (2025): Shaping the future of AI in healthcare through ethics and governance *Relevance:* Sub-Question II (participatory governance) — ethics-first approach emphasizing community engagement and accountability mechanisms.

Paper 7 — Abdul (2025): AI-Driven Ethical Governance *Relevance:* Sub-Question II (adaptive governance) — balance between innovation and accountability across data-intensive sectors including healthcare. Useful for comparative sectoral perspective.

Paper 8 — Kéri (2026): Simulated consciousness in psychiatry AI *Relevance:* Sub-Question III (implementation barriers, safety risks) and Sub-Question II (governance) — unique contribution on psychiatric AI failure modes (simulated empathy, illusion of attachment) not addressed elsewhere.

Paper 9 — Abulibdeh, Celi & Sejdić (2025): Illusion of safety — FDA report *Relevance:* Sub-Question I (evaluation — post-market monitoring gap) and Sub-Question II (regulatory governance) — direct critique of FDA approval process, calls for adaptive community-engaged regulation. Essential counterpoint to Warraich et al.'s FDA perspective.

.3 Search Keyword Themes

See Table .1 for a full listing of keywords used across the five thematic axes for the systematic search.

Table .1: Search keyword thematic axes and associated terms

Thematic Axis	Keywords
AI Evaluation & Validation	AI evaluation, AI validation, AI auditing, algorithmic auditing, model evaluation, AI testing, AI assessment, external validation, real-world validation, deployment evaluation, post-deployment monitoring, continuous evaluation, AI performance assessment, bias detection, robustness testing, AI reliability, explainability, safety evaluation
Participatory Governance	participatory governance, participatory democracy, community participation, stakeholder engagement, citizen participation, deliberative democracy, co-governance, collaborative governance, participatory AI, democratic AI, community-centered AI, public interest technology, algorithmic accountability
Adaptive Regulation	adaptive governance, adaptive regulation, agile regulation, responsive regulation, dynamic regulation, iterative regulation, experimental regulation, learning regulation, regulatory sandboxes, living labs, evidence-based regulation, outcome-based regulation, AI governance, AI oversight, continuous compliance
Evidence & Implementation	evidence-based policy, knowledge translation, research utilization, evidence synthesis, implementation science, knowledge mobilization, actionable evidence, evidence gaps, implementation barriers, policy uptake, research impact, evidence quality
Clinical Health AI	clinical AI, health AI, medical AI, clinical decision support, diagnostic AI, predictive modeling in healthcare, digital health, AI safety in healthcare, clinical validation

.4 Quality Appraisal Rubric

.5 Full Corpus Roster (V3.1, $n = 60$)

The complete roster of the 60 papers comprising the V3.1 corpus, sorted by stance (SPECIFIC_GAP → SUPPORT → ELSE → COUNTER) and then by year (descending). The `Gaps` column shows the binary coding for Gap 1 (ecological validity), Gap 2 (governance capacity), and Gap 3 (pilot trap): 1 = documented, 0 = absent. Stance abbreviations: SG = SPECIFIC_GAP, SUP = SUPPORT, ELS = ELSE, CNT = COUNTER. Confidence abbreviations: hig = high, med = medium, low = low. The full coding matrix (including verbatim evidence quotes, indicator lists, and interaction directions) is available at `https://github.com/Arkoys/ai-health-lit-review` in `analysis/_codings_v3.1_stance.json`.

ID	Year	Src.	Title	Gaps	Stance	Conf.
d:corr/abs-2601-15630	2026	DBLP	Agentic AI Governance and Lifecycle Management . . .	010	SG	low
d:corr/abs-2603-07924	2026	DBLP	Semantic Risk Scoring of Aggregated Metrics: An. . .	010	SG	low
d:npjdm/HusseinZRHHS26	2026	DBLP	Advancing healthcare AI governance through a co. . .	010	SG	low
pmid:41930736	2026	PubMed	Validation of a Risk-Prediction Model in the Pr. . .	100	SG	hig
pmid:41935450	2026	PubMed	Prediction of Early-onset Preeclampsia Using De. . .	101	SG	hig
pmid:41935973	2026	PubMed	Predictive neuroimaging biomarkers of major dep. . .	100	SG	hig
pmid:41936035	2026	PubMed	[Simulated consciousness, pseudo-empathy and th. . .	110	SG	med
pmid:41941973	2026	PubMed	Exploring AI and ML in managing overlap between. . .	101	SG	hig
pmid:41942350	2026	PubMed	From Pilot Trap to Institutional Capacity: A Go. . .	011	SG	hig
pmid:41942817	2026	PubMed	AI Models for Surgical Decision Support in Spon. . .	100	SG	med
ss:5dce643	2026	SemSchol	A comprehensive review of venous thromboembolis. . .	101	SG	hig
ss:65d820b	2026	SemSchol	Beyond binary diagnosis: Key questions on AI ac. . .	101	SG	hig
ss:bf7fd44	2026	SemSchol	Evidence-Driven AI Governance for Healthcare: A. . .	010	SG	hig
d:fllm/QureshiS25	2025	DBLP	Responsible AI to AI Governance: An Awareness-D. . .	010	SG	low
d:ijcnn/Skubis25	2025	DBLP	AI Governance: Ethics of Trustworthy AI in Heal. . .	010	SG	low
d:corr/abs-2512-03098	2025	DBLP	An AI Implementation Science Study to Improve T. . .	001	SG	low
ss:1fe644e	2025	SemSchol	Real-World Evaluation of Large Language Models . . .	110	SG	hig
ss:37453bb	2025	SemSchol	The illusion of safety: A report to the FDA on . . .	110	SG	hig
ss:48816ce	2025	SemSchol	Leadership Challenges in Implementing AI Govern. . .	010	SG	med
ss:5c5299a	2025	SemSchol	AI-based Clinical Decision Support for Primary . . .	001	SG	med
ss:62ec161	2025	SemSchol	Artificial intelligence in diabetology: systema. . .	101	SG	med
ss:644f8df	2025	SemSchol	Preventing unrestricted and unmonitored AI expe. . .	110	SG	hig
ss:79185ce	2025	SemSchol	The Blockchain Deployment Deficit in AI-Enabled. . .	101	SG	hig
ss:89ba9fd	2025	SemSchol	Patient-Reported Outcome Measures in Adults wit. . .	001	SG	med
ss:97341ce	2025	SemSchol	SLaM Image Bank - a real-world diverse London c. . .	100	SG	hig
ss:9b55f76	2025	SemSchol	Artificial intelligence in healthcare: global i. . .	010	SG	hig
ss:9cb5147	2025	SemSchol	Ethical and legal considerations in healthcare . . .	010	SG	med
ss:ca5cf92	2025	SemSchol	Real-World Feasibility of Generative Large Lang. . .	100	SG	hig
ss:d65b399	2025	SemSchol	Ecological Validity Missing in AI-Assisted Clin. . .	100	SG	hig
ss:e676d1d	2025	SemSchol	Governance and Accountability of AI Agents in U. . .	111	SG	hig
ss:eb64145	2025	SemSchol	Deep Learning in Healthcare	100	SG	hig
ss:ffdae9b	2025	SemSchol	Generative AI's healthcare professional role cr. . .	010	SG	hig
ss:27b2cb8	2024	SemSchol	Comparative Governance of AI-Driven Healthcare . . .	011	SG	hig
ss:6fe6e3d	2024	SemSchol	FDA-Approved Artificial Intelligence and Machin. . .	110	SG	hig
ss:7caca59	2024	SemSchol	Navigating regulatory and policy challenges for. . .	110	SG	hig
ss:8fabf24	2024	SemSchol	Shaping the future of AI in healthcare through . . .	010	SG	med
ss:77e1783	2023	SemSchol	From theoretical models to practical deployment. . .	101	SG	hig
ss:ce7982b	2023	SemSchol	Real-World Demonstration of AI to Clinical Cyto. . .	100	SG	hig
2604.11798v1	2026	Other	Budget-Aware Uncertainty for Radiotherapy Segme. . .	100	SUP	hig
d:biostec/ChandradevaM26	2026	DBLP	AI-Based Software as a Medical Device: Bridging. . .	010	SUP	low
d:npjdm/KimHBS26	2026	DBLP	People process technology and operations framew. . .	010	SUP	low
ss:166b63e	2026	SemSchol	NURSE-AI: A Nurse-by-Design Framework for Multi. . .	111	SUP	hig
ss:9444765	2026	SemSchol	Ethical AI Governance in Healthcare CRM Systems	010	SUP	hig
ss:d9f0f2d	2026	SemSchol	AlignInsight: A Three-Layer Framework for Detec. . .	110	SUP	low
d:ijmi/OlawadeWTMBH25	2025	DBLP	Evaluating AI adoption in healthcare: Insights . . .	011	SUP	low
ss:981e769	2025	SemSchol	Post-Marketing Surveillance and Regulation of t. . .	110	SUP	hig
ss:af5ed53	2025	SemSchol	Prioritizing human-AI collaboration in healthca. . .	111	SUP	hig
ss:6ff7c77	2024	SemSchol	Developing a Framework for Self-regulatory Gove. . .	111	SUP	hig
ss:c222df3	2024	SemSchol	FDA Perspective on the Regulation of Artificial. . .	110	SUP	hig
ss:057dc43	2021	SemSchol	Multi-Institution Encrypted Medical Imaging AI . . .	101	SUP	hig
d:bise/BartschBBBBDFJLRSW25	2025	DBLP	Governance of High-Risk AI Systems in Healthcar. . .	000	ELS	low
d:fdgth/FieldsBTJARABAHS25	2025	DBLP	Governance for anti-racist AI in healthcare: in. . .	000	ELS	low
ss:48dc9fe	2025	SemSchol	How AI Transforms Regulatory Submission: Curren. . .	000	ELS	low
ss:6b25670	2025	SemSchol	Generative AI in clinical practice: novel quali. . .	000	ELS	low
ss:edada58	2025	SemSchol	Artificial intelligence in traditional Chinese . . .	000	ELS	med
d:titb/Shaban-NejadMBB21	2021	DBLP	Guest Editorial Explainable AI: Towards Fairnes. . .	000	ELS	low
pmid:41942089	2026	PubMed	A comparative analysis of AI scribes versus hum. . .	000	CNT	med
ss:11855a6	2025	SemSchol	From prediction to action: a retrospective obse. . .	000	CNT	med
ss:6d94421	2025	SemSchol	Prospective Implementation of AI-Assisted CBCT-. . .	000	CNT	med
ss:a64b8f3	2025	SemSchol	Clinical implementation of AI-based screening f. . .	000	CNT	hig

Table .2: Complete V3.1 corpus roster (n = 60), ordered by stance then year (descending). The Gaps column reports the binary coding for Gap 1 (ecological validity), Gap 2 (governance capacity), and Gap 3 (pilot trap). Stance and confidence labels follow the V3.1 coding protocol.

Technical Appendix: AI-Assisted Literature Pipeline

.6 Pipeline Architecture

The literature collection pipeline operates across five functional layers: **data collection**, **paper scoring**, **LLM-based summarization**, **storage and indexing**, and **dissemination**. Figure .1 illustrates the end-to-end flow.

Step 1	`collector.py` queries arXiv, PubMed, Semantic Scholar, DBLP
Step 2	Results deduplicated by DOI / title and passed to `ranker.py`
Step 3	`ranker.py` computes composite priority scores
Step 4	Priority papers passed to `summarizer.py` (LLM)
Step 5	Structured summaries stored in JSON + SQLite
Step 6	Daily digest + weekly synthesis written to `outputs/`
Step 7	Telegram bot delivers digest to registered channel

Figure .1: Literature pipeline execution steps.

The pipeline is implemented as two entry-point scripts: `daily_pipeline.py` (or `daily_pipeline_v2.py`) runs on a scheduled cron job each weekday morning; `weekly_synthesis.py` runs every Friday. Both scripts support a `–push` flag to commit outputs directly to the GitHub repository.

.7 Data Sources and Query Strategy

Four bibliographic sources are queried on each pipeline run:

- **arXiv** (`http://export.arxiv.org/api/query`) — preprints from cs.AI, cs.LG, cs.CL, cs.CY, q-bio, eess.AS, cs.CV. Eight structured queries cover each of the five thematic axes using `all:` field operators. Results parsed from Atom XML response.
- **PubMed** (`https://eutils.ncbi.nlm.nih.gov/entrez/eutils`) — MEDLINE via E-search / E-fetch. Eight PubMed-formatted queries covering clinical AI evaluation, governance, adaptive regulation, evidence synthesis, and participatory engagement.
- **Semantic Scholar** (`https://api.semanticscholar.org/graph/v1`) — free-tier Graph API. Five keyword queries with health-article filtering. Fields requested: paperId, title, authors, abstract, year, venue, externalIds.
- **DBLP** (`https://api.dblp.org/v1`) — targeted venue searches for FAccT, AIES, CHI, ICLR, and NeurIPS proceedings filtered to health-AI-relevant titles.

Query strings are defined in `collector.py` as constants `ARXIV_QUERIES`, `PUBMED_QUERIES`, `SEMANTIC_SCHOLAR_QUERIES`, and `DBLP_VENS`. Deduplication is performed by paper ID and normalized title (80-character window, alphanumeric stripped) against the existing corpus before insertion.

.8 Paper Scoring Algorithm

Each collected paper is scored by `ranker.py` using a five-component composite formula. The total score determines the priority band assigned to the paper, which governs whether it is included in the daily digest and whether an LLM summary is generated.

.8.1 Composite Score Formula

$$\text{Score} = \text{Theme\_Score} + \text{Novelty\_Score} + \text{Recency\_Boost} + \text{Venue\_Score} + \text{Methods\_Bonus}$$

Each component is computed as follows:

- **Theme_Score** — Each of the five thematic keyword groups contributes a weight when its keywords appear in the paper title or abstract: `AI_EVAL_VALID` (weight 2), `PARTICIPATORY_GOV` (weight 2), `ADAPTIVE_REGULATION` (weight 2), `EVIDENCE_IMPLEMENT` (weight 2), `CLINICAL_HEALTH_AI` (weight 1). Pre-tagged themes from the collector are also counted. Theme_Score = matched_themes_count × 2.
- **Novelty_Score** — Detected via linguistic markers in title and abstract. New contribution indicators (*we propose, novel, first to, new framework*) yield 2.0. Extension indicators (*extend, adapt, improve*) yield 1.0. Redundant or incremental work yields 0.5.

- **Recency_Boost** — Based on days since publication date: ≤30 days = 2.0; ≤90 days = 1.0; ≤365 days = 0.5.
- **Venue_Score** — Top-tier venues (FAccT, AIES, NeurIPS, ICML, ICLR, Nature Medicine, JAMA, Lancet, Science, Medical Internet Research, Implementation Science) yield 3.0. Recognized venues (Frontiers, BMJ Global Health, Nature Biomedical Engineering, JAMA Network Open, IEEE J Biomed Health Inform) yield 1.5. Preprints (arXiv, medRxiv, bioRxiv) yield 0.5.
- **Methods_Bonus** — Empirical/lab studies (RCT, prospective cohort, validation study) yield 2.0. Systematic reviews and meta-analyses yield 1.5. Frameworks and conceptual proposals yield 1.0. Commentary and editorials yield 0.5.

.8.2 *Priority Bands*

Table .3: Priority band thresholds and pipeline behaviour.

Priority	Score range	Pipeline behaviour
3 (Must Read)	≥ 7.0	Daily digest highlight; LLM summary generated
2 (Important)	4.0 – 6.9	Daily digest listing; no LLM summary
1 (Supplementary)	1.5 – 3.9	Database only
0 (Low)	< 1.5	Database only; excluded from outputs

The full scoring implementation is in `ranker.py`, functions `score_paper()` and `rank_papers()`.

.9 Database Schema

Papers are persisted in two formats: a JSON document store (`data/papers.json`) and a SQLite database (`data/papers.db`). The JSON store holds the canonical scored corpus used by digest generation. The SQLite database (defined in `database.py`) supports structured queries and is initialised with the following schema:

```
CREATE TABLE papers (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    paper_id          TEXT UNIQUE NOT NULL,
    source            TEXT NOT NULL,
    title             TEXT NOT NULL,
    authors           TEXT,
    abstract          TEXT,
    url               TEXT,
    pdf_url           TEXT,
    published_date    DATE,
    venue             TEXT,
    doi               TEXT,
    keywords          TEXT,          -- JSON array
    score             REAL DEFAULT 0.0,
    priority          INTEGER DEFAULT 0,
    summary           TEXT,
    critique          TEXT,
    methods           TEXT,
    gaps              TEXT,
    related_papers    TEXT,          -- JSON array
    processing_status TEXT DEFAULT 'pending',
    created_at        TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_updated      TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE daily_digests (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    digest_date       DATE NOT NULL UNIQUE,
    papers_included   TEXT,          -- JSON array of paper_ids
    total_papers_count INTEGER,
    highlight_papers  TEXT,          -- JSON array of top papers
```

```
    trends_observed   TEXT,
    gaps_identified   TEXT,
    generated_at      TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    sent_to_telegram  BOOLEAN DEFAULT 0,
    sent_to_email     BOOLEAN DEFAULT 0,
    sent_to_gdocs     BOOLEAN DEFAULT 0
);
```

Indexes are created on `score`, `published_date`, and `processing_status`.

.10 Output Products

Three classes of output are produced by the pipeline:

- **Daily digests** (`outputs/digests/digest_YYYY-MM-DD.md`) — Markdown summaries of all papers collected in a given day, organized by priority. Includes title, authors, source, abstract excerpt, thematic tags, and composite score. If a digest already exists for the same date, the pipeline creates a versioned file (`digest_YYYY-MM-DD_v2.md`, etc.) rather than overwriting.
- **Weekly synthesis reports** (`outputs/weekly/weekly_YYYY-MM-DD.md`) — Aggregates and cross-references papers collected over the preceding seven days. Includes a re-ranked list of the top papers, a thematic distribution table, and a narrative synthesis of emerging trends.
- **Topic-trend matrices** (`outputs/topic_trends/matrix_YYYY-MM-DD.md`) — A 4-week rolling window matrix showing the proportion of papers assigned to each of the five thematic axes over time, enabling identification of shifts in the literature's thematic focus.

All outputs are committed to the GitHub repository (`Arkoys/ai-health-lit-review`) using a Fine-Grained Personal Access Token with read-write access to that repository.

Code Availability

Full pipeline source code (Python scripts, scoring algorithm, database schema, daily pipeline, and synthesis) is publicly available at the project repository: `https://github.com/Arkoys/ai-health-lit-review`. The version of the code corresponding to the findings reported in this review is archived at the commit referenced in the Section 2 methods note. Pipeline scripts are anonymized: deployment credentials, messaging tokens, and scheduling configuration are managed in external environment files and are not part of the public repository.